

PERFORMANCE MARKETING AND HEALTH OUTCOMES IN MEDICAL CROWDFUNDING

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Abstract

Digital platforms routinely allocate marketing resources using short-term performance metrics, such as clicks or conversions, that capture consumer preference. On crowdfunding platforms, each conversion contributes to the fundraising campaign of an identifiable beneficiary and, in turn, shapes downstream outcomes for that beneficiary. Whether allocating resources based on what the paying side rewards also maximizes impact on beneficiary outcomes is therefore an empirical question. We study this question in the context of medical crowdfunding and find that performance-based marketing concentrates spend on campaigns with the best donation performance rather than those where timely fundraising most accelerates treatment and improves patient health. To quantify the effect, we develop a sequential framework, the marketing-to-health (M2H) chain, which links marketing spend to fundraising progress, fundraising to treatment timing, and treatment timing to patient health, and combines these links into the Health Return on Marketing (HROM), a marginal health return metric to guide marketing allocation decisions. Using proprietary daily campaign-level data and quasi-experimental variation in advertising exposure, we estimate each link of the chain and simulate marketing allocation policies that take into account both donation performance and health outcomes. Donation-maximizing allocation is substantially misaligned with health impact: health-aware reallocation improves patient health by up to 14.6% relative to the donation-maximizing policy. Because the donation–health frontier is concave, platform managers need not abandon fundraising performance to improve patient health: a policy that trades off only 2.3% of donations improves patient health by 9.1%. Incorporating outcome-based returns into marketing allocation can thus meaningfully improve real-world outcomes whenever short-term consumer responses differ in their downstream impact.

Keywords: Resource Allocation; Medical Crowdfunding; Health Outcomes; Performance Marketing; Digital Marketing.

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1. INTRODUCTION

Donation-based medical crowdfunding platforms such as GoFundMe, Ketto, and Milaap have grown rapidly across low-, middle-, and high-income countries and have become an important source of healthcare financing in markets with incomplete insurance coverage¹ (Cai et al., 2025). These platforms connect patients facing acute medical expenses with dispersed donors. Unlike traditional charity, where donations are collected for a general group of recipients, each donation in medical crowdfunding directly contributes to the fundraising campaign of an identifiable patient and, in turn, affects the downstream outcomes for the patient beneficiary (Belleflamme et al., 2015; Burtch and Chan, 2019). A key strategic marketing decision crowdfunding platforms make is to allocate scarce promotional exposure across many simultaneously active campaigns (Song et al. 2022; Impact Guru 2026).

Digital platforms typically rely on short-term performance metrics capturing consumer preference, such as clicks or conversions, to allocate exposure that maximizes return on marketing. In the context of a medical crowdfunding platform, performance is typically measured in terms of donations, which capture donor preference and what donors find salient. Moreover, most such platforms are for-profit entities whose revenue derives from fees or tips on donations (Kenworthy 2019), so allocating exposure that maximizes donations also maximizes platform revenue. However, relying on donation performance to allocate marketing resources creates the potential for a fundamental misalignment because platform marketing affects not only which campaigns receive attention, but also which patients gain access to treatment. Donor behavior is shaped by identifiable-victim effects, social proximity, and framing rather than

¹ In many low- and lower-middle-income countries, out-of-pocket payments remain the main source of health financing, often exceeding half of total health spending (World Health Organization 2024).

medical severity (Small et al., 2007; Kogut and Ritov, 2005; Sudhir et al., 2016; Hur et al., 2023). Thus, campaigns that are most effective at attracting donations are not necessarily those for which earlier funding would most accelerate treatment and improve health outcomes. In this paper, we ask whether allocating marketing resources based on donation performance is misaligned with maximizing impact on beneficiary outcomes. Furthermore, if the misalignment is substantial, it raises a central marketing allocation problem for platform managers: how should donation-based crowdfunding platforms allocate marketing resources when donors' revealed preference is misaligned with impact on beneficiary outcomes?

These questions are relevant not only from the perspective of a platform with a social mission but also for platform design. A central theme in platform design is how platforms should make strategic decisions that account for outcomes on both sides of the market as well as the platform's ability to generate revenue (Belleflamme and Peitz 2021). Donation-based crowdfunding platforms often have incentives to value beneficiary outcomes for reasons beyond their social mission because successful project outcomes and feedback about downstream beneficiary impact makes donors feel more emotionally rewarded (Aknin et al. 2013), encourage donor return and improve repeat participation (Althoff & Leskovec 2015). In practice, donation-based medical crowdfunding platforms actively curate and regularly publish patient success stories as a core donor-engagement and retention strategy (e.g., Milaap, 2024; Ketto, 2023). Answering these questions empirically requires going beyond the immediate consumer (in crowdfunding, donors) responses to platform's strategic decisions. Marketing research has often treated consumer search, choice, purchase, or contribution as the endpoint of analysis, leaving the downstream consequences of those decisions less understood. Accordingly, Moorman et al. (2021) call for research linking marketing-related search and purchase decisions to actual health outcomes, while Madan et al. (2023) note that charitable-giving research has focused primarily

on what drives consumers to donate rather than what happens after a contribution is made. However, medical crowdfunding platforms typically do not have direct access to patients' clinical records, which are maintained by healthcare providers. These records provide precise information on diagnoses, treatment timing, and clinical severity, but are generated across fragmented provider systems, and are rarely interoperable with platform data. Moreover, routine clinical records may not systematically capture patients' experiences during recovery, including pain, mobility, daily functioning, and quality of life, much of which unfolds outside hospital encounters. Platforms must therefore rely largely on patient- or campaigner-provided updates which contain rich descriptions of patients' experience during recovery but are reported intermittently and in unstructured form.

We address this gap by developing a sequential framework, the marketing-to-health (M2H) chain, that links platform marketing decisions to patient health outcomes. The framework chains together individual links connecting promotional spending to campaign's timely fundraising progress, fundraising progress to treatment timing, and treatment timing to the patient's health trajectory. We combine these links to construct Health Return on Marketing (HROM), a measure of the marginal improvement in cumulative patient health generated by marketing spend. This measure transforms downstream health outcomes into a marginal health return metric that can be used directly in allocation decisions.²

Operationalizing the M2H framework poses several measurement and estimation challenges because the data typically observed by platforms do not map directly to the links in the chain. Platforms typically possess detailed data on marketing resource allocation and donation flows in real time. However, donations reflect both organic appeal and the impact of platform marketing,

² Section 3.1 (Model: Overview) describes the construction of the cumulative patient health measure, and discusses its relationship to health benefit measures used in health economics for scarce resource allocation.

while platforms may allocate more resources to campaigns expected to perform well. Observed donation performance therefore confounds inherent appeal with marketing-driven lift. Information farther downstream from donations is also less directly observed: treatment milestones and recovery are often reported intermittently by campaigners through unstructured updates, making treatment timing and health status imperfectly measured. Moreover, the urgency of treatment may shape the fundraising progress, making it difficult to isolate how additional funding affects treatment timing and patient health trajectory. Platforms must therefore address three general empirical challenges: identify the causal effect of marketing on fundraising progress, convert downstream treatment and health information into quantitative measures, and estimate how fundraising progress affects treatment timing and patient health.

We implement this framework while addressing these empirical challenges using novel, proprietary data from a medical crowdfunding platform to examine the research questions posed earlier:

- (i) Is the misalignment, as measured by the loss in patient health from performance marketing, substantial?
- (ii) How much patient health can be recovered through alternative marketing resource allocation policies and at what cost to donation performance?

Our empirical setting is one of the largest medical crowdfunding platforms in India. The platform connects patients, primarily based in India and facing acute medical expenses for major surgical procedures, with donors in India and across the globe. Between 2020 and 2024, the platform hosted more than 100,000 campaigns and processed over 4M individual donations. In this setting, we collect rich operational data recording daily campaign-level marketing expenditures by the platform, time-stamped donations from individual donors, and patient-reported updates tracking treatment milestones and recovery. Our analysis focuses on 1,280

hospital-managed campaigns spanning seven clinical pathways and involving major invasive surgical procedures, for which treatment timing and health returns can be credibly linked to early fundraising dynamics.

We address the empirical challenges in estimating the links in the M2H framework in our setting as follows. First, to identify the causal effect of marketing on donations, we exploit two sources of exogenous variation unique to our setting. The first is an operational feature: campaigns experience variable delays between the start of fundraising and the onset of paid marketing because of operational processes such as creative production, compliance review, and hospital coordination. These campaign-specific delays generate quasi-random variation in the timing of marketing exposure. The second is a rare and salient regulatory shock: a 2023 regulatory restriction on charitable-cause advertising that effectively eliminated YouTube/Google as a marketing channel, sharply constraining the platform's allocation choices. Second, we address the challenge of converting patient-reported downstream outcomes (patient treatment milestones and health status) into quantitative measures that can be used to estimate the M2H chain by using a large language model. For each patient update, the LLM assigns a single health score on a 1–5 scale by synthesizing information on patient's functional status, mobility, pain, care dependency and overall quality of life. We validate the scoring protocol through an independent rescoring exercise using an alternative model that yields near-perfect agreement. Third, to identify the effect of fundraising progress on treatment timing, we leverage an institutional feature that generates plausibly exogenous variation in the funding–timing relationship: hospitals do not schedule procedures until a minimum funding threshold is met. Because treatment timing and health are observed only intermittently through campaign updates, we recover latent treatment dates and health trajectories using a joint model of treatment timing and latent health that allows us to estimate the full M2H chain.

Our findings yield several insights. First, the misalignment is substantial: relative to a donation-maximizing policy, a budget-neutral health-maximizing marketing allocation improves patient health by up to 14.6%, indicating that performance-marketing tends to direct promotional resources away from campaigns where marginal funding progress generates the largest health gains. Second, the donation–health frontier is concave implying that much of this health loss can be recovered at little cost to fundraising performance. A budget-neutral health-aware policy that reallocates marketing spend across concurrently active campaigns, while accounting for both donation performance and health return, can improve patient health by up to 9.1%. by trading off merely 2.3% of donation performance, vis-à-vis the donation-maximizing policy. Third, the health gains are highly asymmetric. Campaigns that gain marketing spend improve in health three to ten times more than the losses incurred by campaigns that lose marketing spend, a pattern that holds across every clinical pathway. Health-aware allocation therefore directs resources toward underfunded campaigns with high health returns and reduces concentration in fundraising progress, even without explicitly targeting measures of need or optimizing an equity objective.

Our study contributes to three literature streams. First, to research on the impact of marketing on society, we move beyond immediate consumer responses by tracing how platform marketing and donation outcomes affect treatment timing and patient health. Second, to platform-design research, we show that allocation policies based on donor-side performance can be substantially misaligned with impact on downstream beneficiary outcomes, highlighting the importance of accounting for outcomes on both sides of the platform. Third, to marketing resource-allocation research, we develop the Health Return on Marketing and propose health-aware allocation policies that jointly account for donation performance and patient health, yielding substantial health gains with minimal loss in fundraising performance.

Our study also contributes to practice by providing platforms with a measurable and implementable basis for health-aware allocation. HROM can be computed from campaign characteristics available at launch, allowing platforms to incorporate patient health into real-time marketing decisions. Managers need not abandon fundraising performance: donation-retention constraints identify reallocations that improve patient health while preserving nearly all the donations generated under a donation-maximizing policy.

The remainder of the paper is organized as follows. Section 2 reviews related literature and develops the conceptual framework linking marketing allocation to health outcomes. Section 3 presents the model in three parts: the three-link chain from marketing spend to patient health, the HROM metric that integrates them, and the counterfactual framework for evaluating the tradeoff between fundraising performance and patient health. Section 4 describes the institutional setting, data sources, and descriptive patterns. Section 5 details estimation and identification. Section 6 presents the results.

2. LITERATURE POSITIONING AND CONCEPTUAL BACKGROUND

We conceptualize medical crowdfunding as an allocation problem in which platforms distribute scarce marketing resources to direct promotional exposure across campaigns using performance metrics that may be systematically misaligned with the downstream impact those resources ultimately produce. Understanding how such allocation decisions propagate to patient outcomes requires integrating two strands of literature: (i) research on how marketing and campaign characteristics shape donor behavior, and (ii) research on marketing resource allocation in settings with competing objectives and heterogeneous returns.

Marketing Effects on Donor Behavior and Downstream Outcomes

A large body of work shows that crowdfunding outcomes depend heavily on donor psychology and campaign presentation rather than on underlying need or impact. Donors respond strongly to identifiability and salience: a single, vividly presented beneficiary attracts more giving than diffuse or statistical need (Small et al., 2007; Kogut and Ritov, 2005), and contributions depend sensitively on which case attributes are highlighted (Hur et al., 2023). Emotional responses to campaign narratives and imagery further increase donations independently of medical severity (Sudhir et al., 2016; Yazdani et al., 2025). Giving also exhibits scope insensitivity, with contributions failing to scale proportionally with the magnitude or prevalence of need (Slovic, 2007). Platform design choices—such as the degree of donor control over targeting—shape giving patterns as well, often raising fundraising without improving the alignment between funding and need (Esterzon et al., 2023; Burtch et al., 2015). These findings imply that donation flows are governed by response functions only weakly—and often negatively—related to underlying need or marginal impact. Donor-driven funding is therefore unlikely to reflect health-maximizing priorities, and at the population level these patterns track and can widen disparities in access to care (Kenworthy and Igra, 2022).

A separate literature establishes that marketing does not merely move transactions but can shift real health behavior and outcomes: targeted outreach raises cancer-screening take-up (Chen et al., 2020), and direct-to-consumer pharmaceutical advertising generates spillovers onto reproductive-health outcomes (Kim and KC, 2020). In crowdfunding, the analogous consequences run through funding itself—marketing determines which campaigns raise money, and money determines which patients reach treatment—yet whether the allocation of promotional exposure shapes patient health has not been examined.

These behavioral regularities translate into a platform-level distortion through two structural features of the setting. First, donations are earmarked³: funds are committed to the campaign that attracts them and cannot be reallocated to uses where the same resources would generate greater impact. Second, platforms allocate exposure using performance-based signals derived from donor responsiveness. Jointly, these features contribute to a persistent misalignment: campaigns that attract donations most readily receive more exposure, which generates more earmarked funding, reinforcing their advantage regardless of health impact. When donor appeal and marginal health return diverge across campaigns, performance-based allocation concentrates both attention and funding on donor-attractive campaigns. The empirical extent and patient health consequences of this misalignment remain unknown.

Marketing Resource Allocation

The problem of allocating marketing resources has a long tradition in the literature. Prior work has examined allocation across customers using purchase behavior and lifetime value (Reinartz et al., 2005; Kumar et al., 2008), across products or regions using aggregate response models (Fischer et al., 2011), and over time in the presence of dynamic advertising effects (Naik and Raman, 2003; Dubé et al., 2005). A common feature of this literature is that allocation is optimized with respect to a single firm-side objective—revenue, profit, or return on marketing investment—that is either directly observed or closely proxied by contemporaneous outcomes.

More recent work on platforms and recommendation systems recognizes that allocation decisions often affect multiple stakeholders with potentially conflicting objectives.

³Earmarking of donations to specific recipients is a defining feature of crowdfunding and widespread in charitable fundraising more broadly; see Fuchs, de Jong, and Schreier (2020) for cross-cultural evidence on its appeal to donors and Joireman et al. (2020) for evidence on donor reactions to redirected donations. Our analysis takes earmarking as given and focuses on the allocation of marketing spend, which the platform controls directly, rather than the reallocation of donations across campaigns.

Recommendation policies that maximize firm revenue can reduce consumer surplus or distort exposure (Hosanagar et al. 2008; Zhang et al. 2021), and multi-objective frameworks have been developed to account explicitly for such trade-offs (Wang et al. 2024). In crowdfunding, recommendation policies can produce highly unequal funding distributions, concentrating contributions on a small subset of campaigns (Song et al. 2022).

Our setting fits squarely in the budget-allocation tradition in its mechanics: the platform distributes a centralized marketing budget across simultaneously active campaigns through third-party ad networks, and the decision variable is how much exposure each campaign receives, not which individual donor sees which campaign. What distinguishes it is the objective. Existing frameworks typically assume the objective is directly observed or well proxied by immediate performance; here, the platform optimizes on observed donation response, while the outcome that matters—patient health—unfolds over time and depends on how funding converts into access to care. The setting thus has three defining features: (i) earmarked flows, which bind funds to the activities that generate them; (ii) centralized exposure control, which gives the platform discretion over allocation; and (iii) health consequences that are realized later and unobserved when allocation decisions are made. When the correlation between the intermediate signal and the eventual outcome is weak or negative, optimizing on observed response systematically selects against high-impact activities.

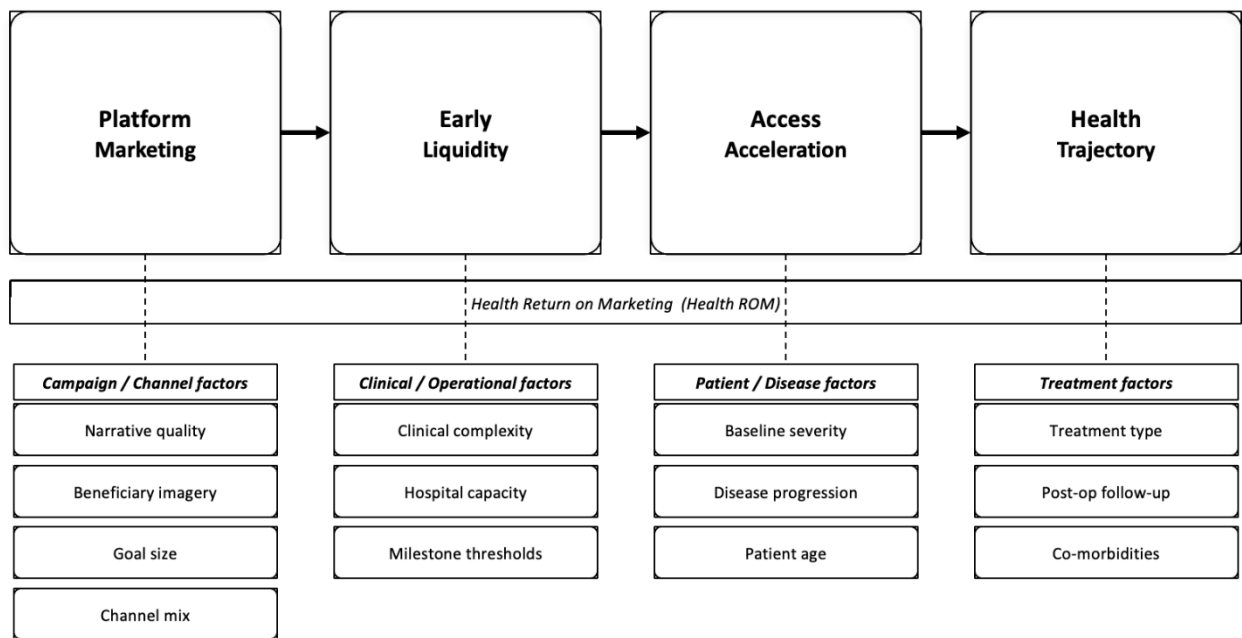
We address this problem by modeling the mapping from marketing allocation to patient outcomes and constructing a return metric that takes into account these delayed effects, allowing allocation rules to be evaluated even when the objective is not directly observed but must be inferred through a multi-stage process linking marketing inputs to downstream outcomes.

A Multi-Stage Framework Linking Marketing Allocation to Patient Health

To formalize this problem, we develop a multi-stage framework that links platform marketing expenditure to patient health through three sequential mechanisms, illustrated in Figure 1. The framework reflects the structural feature noted above: because donations are earmarked, marketing expenditure does not simply generate revenue but routes both funding and its health consequences to the campaigns that attract donors. Evaluating allocation decisions therefore requires tracing the effect of marketing through each stage of this process.

We integrate these components to construct the Health Return on Marketing (HROM), defined as the marginal improvement in patient health—measured as cumulative health over a recovery horizon—per unit of marketing expenditure. HROM internalizes the full causal chain from marketing to donations, from funding to treatment access, and from treatment timing to health outcomes. It thus provides a forward-looking, decision-relevant metric that converts an otherwise myopic allocation problem into one evaluated in terms of patient health.

Figure 1: A Multi-Stage Framework Linking Platform Marketing to Patient Health



Notes: The boxes represent the three sequential stages linking platform marketing to patient health. Arrows indicate causal flow. Dashed lines connect each stage to its primary moderating factors (bottom rows). "Health ROM" denotes the Health Return on Marketing (HROM), the integrated health measure spanning all three links.

Marketing → Early Liquidity. Marketing expenditure increases awareness and generates donations according to campaign-specific response functions. The timing of these donations is critical: early funding relaxes liquidity constraints that delay treatment initiation. The strength of this link depends on campaign characteristics such as narrative quality, goal size, and beneficiary presentation, as well as on channel-specific factors that affect reach and conversion efficiency.

Early Liquidity → Access Acceleration. When insurance is absent or insufficient, as is the case in many lower- and middle-income countries, treatment is contingent on the availability of funds; providers do not schedule procedures until financial commitments are secured, making early liquidity a direct determinant of treatment access. As a result, increases in early funding can produce discrete reductions in time to treatment, particularly for procedures requiring advance financial commitment or multiple preparatory stages. Clinical complexity, hospital capacity constraints, and the structure of funding thresholds shape the magnitude and nonlinearity of this relationship.

Access Acceleration → Health Trajectory. Earlier treatment improves patient outcomes by shifting the timing of recovery. The health gain from earlier access arises because patients spend more time in higher health states within a given horizon. The magnitude of this gain varies across conditions: treatments with large and rapid effects exhibit higher returns to earlier access, while conditions with gradual recovery generate smaller but positive gains. Patient characteristics, disease progression, and treatment protocols further moderate the relationship between timing and health outcomes.

This decomposition provides a decision-relevant foundation for allocation under misaligned objectives. By mapping marketing inputs to patient health through a sequence of steps,

the framework enables comparison of alternative allocation rules—performance-based or HROM-based—under fixed resource constraints and identifies how heterogeneity across campaigns translates into differences in marginal patient health returns.

3. MODELING THE BUDGET ALLOCATION AND HEALTH RETURN ON MARKETING

In this section, we formalize the three-link framework developed above. We specify each link as an estimable model, combine them into the HROM metric, and use HROM to characterize the budget allocation problem that traces the tradeoff between fundraising performance and patient health. We begin with an overview of the allocation setting before formalizing each link in turn.

3.1 Overview

Each day t , the platform allocates a fixed marketing budget B_t across a portfolio of simultaneously active campaigns (defined as campaigns within a fixed time window t' days from campaign start), choosing non-negative spend levels s_i that exhaust the budget. Its current rule is performance-driven: campaigns that generated more donations in recent days receive disproportionately more spend going forward. This rule maximizes near-term fundraising—but, as previously argued, fundraising efficiency need not coincide with where marketing most improves patient health.

To make that distinction operational, we define the Health Return on Marketing (HROM) as the marginal gain in cumulative patient health per MU of marketing spend. HROM traces the full causal chain—from spend to donations, donations to treatment timing, and treatment timing to health—so that marketing resource allocation can be evaluated by its downstream health consequences rather than by donation response alone.

Estimating HROM requires quantifying how a marginal MU of spend propagates through three sequential stages, which we decompose as:

$$\text{HROM}_{i,t} \equiv \frac{\partial \text{AUC}_i}{\partial S_{i,t}} = \underbrace{\frac{\partial D_{i,t}}{\partial S_{i,t}} \times \frac{\partial F_{i,\tau}}{\partial D_{i,t}}}_{\text{Link 1: Marketing} \rightarrow \text{Liquidity}} \times \underbrace{\frac{\partial T_i}{\partial F_{i,\tau}}}_{\text{Link 2: Liquidity} \rightarrow \text{Access}} \times \underbrace{\frac{\partial \text{AUC}_i}{\partial T_i}}_{\text{Link 3: Access} \rightarrow \text{Health}}, \quad (1)$$

where $S_{i,t} \geq 0$ denotes daily marketing spend for campaign i on day t , and $D_{i,t}$ denotes the resulting daily donation amount. Donations accumulate deterministically into funding progress: $F_{i,\tau} = \sum_{t=1}^{\tau} D_{i,t} / \text{goal}_i$ is the fraction of the fundraising goal reached by the end of the early window of τ days. Treatment timing T_i is the time from campaign launch to the treatment milestone, and AUC_i is the area under the patient's latent health trajectory over a fixed horizon, analogous to the quality-adjusted life year (QALY), a generic measure of the value of health outcomes, widely used in health policy and clinical decision-making (Pliskin et al. 1980; Weinstein et al. 2009). The three bracketed terms correspond directly to the links of our proposed M2H framework: marketing to liquidity (Link 1), liquidity to access (Link 2), and access to health (Link 3). The remainder of this section develops each link in turn and then specifies how the platform's budget is allocated under alternative policies.

3.2 Link 1: Marketing $\rightarrow$ Liquidity

The first link quantifies how platform-allocated marketing spend converts into donations. For each campaign-day, the platform assigns marketing expenditure, and donors respond with a flow of donations, $D_{i,t} \geq 0$. In addition to marketing inputs, donation dynamics are also affected by time-varying confounders—campaign momentum, campaign lifecycle growth and decay patterns, and progressive saturation of the campaign's donor pool—so the model must account for these confounders.

Following the standard constant-elasticity (multiplicative) specification for non-negative outcomes (e.g., Santos Silva and Tenreyro, 2006), we write daily donations as an exponential conditional mean with an additive error

$$D_{it} = \exp\{\beta_{S_1,i}\ln(1 + S_{it}) + \gamma_{D_1}\ln(1 + D_{i,t-1}) + \dots + \gamma_{D_2}\ln(1 + \text{cum}D_{i,t-1}) + \beta_{S_2}\ln(1 + \text{cum}S_{i,t-1}) + \alpha'X_i\} + \varepsilon_{it}. \quad (2)$$

where S_{it} denotes total marketing expenditure for campaign i on day t . Because the exponential index is the conditional mean and spend enters in logs, $\beta_{S_1,i}$ is the elasticity of expected donations with respect to $1 + S_{it}$.

Equation (2) also controls for lagged donations, $\ln(1 + D_{i,t-1})$, and cumulative donations, $\ln(1 + \text{cum}D_{i,t-1})$. The former controls for time-varying campaign-specific unobservables which may be serially correlated such as, word of mouth about the campaign or general inertia in donation behavior, while the latter controls for the diminishing marginal likelihood of donors contributing to a campaign that others have already funded (Burtch et al. 2013). The specification also controls for cumulative past spend, $\ln(1 + \text{cum}S_{i,t-1})$, which captures within-campaign saturation of marketing as repeated exposure accumulates (Naik et al. 1998). Finally, the vector X_i includes fixed effects observed at campaign level: hospital, clinical pathway, patient age group, as well as time-varying covariates: days since campaign start, pre-activation donation rate, day-0 organic donations, and campaign goal and indicators for the spend-activation day and campaign-days post activation.

Heterogeneity in the marketing spend elasticity of donations $\beta_{S_1,i}$, is central to our mechanism. We introduce observed heterogeneity, by segment $s(i)$, defined by the campaign's clinical pathway group and patient age group, so the spend-to-donations elasticity $\beta_{S_1,i}$ varies with the medical and demographic characteristics of the beneficiary population. A large body of

evidence on identifiable-victim and sympathy effects shows that donors respond more strongly to some beneficiaries—young children, or particular clinical conditions—than to others of comparable medical need (Small et al. 2007; Kogut and Ritov 2005; Sudhir et al. 2016; Yazdani et al. 2025). Segments whose beneficiaries are more donor-attractive convert a marginal MU of marketing into donations more efficiently, and the segment-specific elasticity $\beta_{S_1,i}$ allows for this variation. To the extent that performance-based allocation channels disproportionate spend toward these high-conversion segments, it does so regardless of where early funding would most improve patient health.

Because the platform allocates spend toward campaigns with strong recent performance and unobserved donor appeal, marketing spend is endogenous: ordinary estimates of $\beta_{S_1,i}$ would conflate the productivity of marketing with the platform’s tendency to fund campaigns that are already attracting donors. We recover the causal elasticities, which are central to obtain campaign-level counterfactual donation performance under simulated alternative marketing allocation policies, using a control-function approach built on two instruments—campaign activation delays and the 2023 Google/YouTube advertising restriction—described in Section 5.1.

3.3 Link 2: Early Liquidity to Treatment Access

The second link captures how early-window fundraising accelerates access to treatment. Let T_i denote the number of days from campaign launch to the first qualifying treatment milestone for patient i (e.g., surgery, transplant, or other major invasive procedure). Because the platform does not observe the exact date of treatment milestone—and instead receives updates only when caregivers report them after the milestone—the true event time lies between the last update indicating treatment was still pending and the first update indicating that it was completed. In these cases, treatment timing is identified to an interval, while campaigns that never report treatment

before the end of the observation period are right-censored. This motivates the use of a censoring-aware duration model (Prentice and Kalbfleisch, 1979; Kalbfleisch and Prentice, 2002; Sun, 2006).

We model time to treatment using a log-normal Accelerated Failure Time (AFT) specification (Prentice and Kalbfleisch, 1979; Chintagunta and Dong, 2006). The AFT form parameterizes the timing of treatment, $\log(T_i)$, directly as a linear function of covariates, which yields a direct time-shift interpretation: a unit change in a covariate translates into a proportional acceleration or deceleration of the time to treatment. This interpretation fits our setting, where the quantity of interest is how additional early fundraising accelerates the timing of treatment and, through this timing channel, increases the likelihood that treatment is observed within the sample window.

Progress toward the goal often relaxes binding financial constraints required for hospitals to schedule treatment, whereas funds raised beyond the goal have diminished marginal impact. To capture these differential effects for funds received above and below the fundraising goal, we allow for a linear spline with a knot at $F_{i\tau} = 1$, with $F_{i\tau}$ denoting the share of the fundraising goal raised in the first τ days.

Considering this, we defined the AFT model as

$$\log(T_i) = \alpha_0 + \beta_{F_1 i} F_{i\tau} + \beta_{F_2 i} (F_{i\tau} - 1)^+ + \beta_G G_i + \beta_M M_i + \phi_{h(i)} + e_i, \quad (3)$$

where $(x)^+ = \max(0, x)$. The goal variable G_i accounts for the possibility that campaigns with larger financial requirements—typically reflecting more complex or resource-intensive procedures—have longer baseline treatment delays. The covariate M_i is the platform’s own matching contribution as a share of early donations, with coefficient β_M capturing any independent effect of the platform’s co-investment on treatment timing. The hospital random intercept $\phi_{h(i)}$ absorbs persistent differences in scheduling capacity, operational delays, and institutional practices

across hospitals. Finally, the disturbance term $e_i \sim N(0, \sigma^2)$ captures residual unobserved heterogeneity in treatment timing beyond observed funding progress, clinical pathway, campaign characteristics, and hospital effects.

Heterogeneity in the funding sensitivity of treatment timing, β_{F_1i} , is central to this link. As in Link 1, we allow it to vary by *clinical pathways* x *patient groups* segments to capture differences in how funding progress translates into treatment access across clinical pathways and patient groups, partially pooling across the segments through a hierarchical structure,

$$\beta_{Fi} \sim N(\mu_{1s(i)}, \sigma_{1s(i)}^2), \quad s = 1, \dots, S. \quad (4)$$

While the AFT specification models time to treatment rather than the binary occurrence of treatment, it also accommodates the fact that observed treatment completion depends on timing relative to the observation window. Campaigns that do not report treatment are treated as right-censored, reflecting that treatment may occur after the end of the sample period rather than not occurring at all. As a result, differences in early fundraising affect not only the expected timing of treatment but also the likelihood that treatment is observed within the data window. In particular, campaigns with lower early fundraising tend to exhibit longer treatment times, making treatment less likely to be observed before censoring.

3.4 Link 3: Treatment Timing to Health Trajectory

The third link describes how a patient's health evolves over time and how treatment timing translates into cumulative health outcomes. The object of interest in this link is how delays in treatment generate health losses through prolonged exposure to poor health states. We therefore require a model that captures (i) stagnation prior to treatment at a low health level, (ii) a treatment-induced improvement, and (iii) post-treatment stabilization toward a new health level.

Let $h_i(t)$ denote patient i 's latent health at time t . Health status is observed only through irregularly spaced update reports collected during platform-initiated follow-up calls, which reflect noisy information about symptoms, mobility, discomfort, and recovery. Following the health econometrics literature on self-reported health as a noisy measure of latent health status (Michaud and van Soest 2008; Van den Berg et al. 2011), we model health as a latent continuous-time process observed at irregular intervals with measurement error.

We expect treatment at time T_i to induce a discrete and persistent improvement in the level of the health process. Prior to treatment, health typically remains flat or gradually declines and after treatment, health improves and then converges toward a new long-run equilibrium. We model this mechanism as a shift in the equilibrium level governing the health process rather than as a purely transitory shock. We provide model-free empirical evidence consistent with this pattern in Section 4 (Figure 4).

To capture these features, we represent the latent health path as an Ornstein–Uhlenbeck (OU) process with a one-time shift in its long-run equilibrium at the treatment event. The OU process is analytically tractable under irregular observation spacing (Durbin and Koopman 2012), has been widely used in finance to model mean-reverting dynamics (Vasicek 1977), and in marketing to represent continuous-time latent states such as advertising goodwill (Naik et al. 1998; Jap and Naik 2008). Its mean-reverting structure naturally captures both the persistence of pre-treatment health conditions and the gradual stabilization of post-treatment recovery.

Conditioning on treatment time T_i , the expected health trajectory evolves as:

$$\mathbb{E}[h_i(t) | T_i] = \begin{cases} \mu_{\text{pre},i} + e^{-\kappa_i t} (h_i(0) - \mu_{\text{pre},i}), & t < T_i, \\ \mu_{\text{post},i} + e^{-\kappa_i (t-T_i)} (h_i(T_i^-) - \mu_{\text{post},i}), & t \geq T_i, \end{cases} \quad (5)$$

where $\mu_{\text{pre},i}$ denotes the pre-treatment equilibrium, $\mu_{\text{post},i} = \mu_{\text{pre},i} + J_i$ is the post-treatment equilibrium, J_i captures the treatment gain, and κ_i is the mean-reversion speed. This formulation highlights that treatment operates through a discrete upward shift in the long-run health level, with realized health converging gradually toward that higher level at rate κ_i .

Heterogeneity in the treatment-induced health gain, J_i , is central to this link. As in Links 1 and 2, we allow it to vary by *clinical pathways* x *patient groups* segments. The pre-treatment equilibrium and treatment gain are modeled as:

$$\begin{aligned}\mu_{\text{pre},i} &= \beta_{\mu,0} + \beta_{\mu,s(i)} + \beta_{\mu,g} g_i \\ J_i &= \beta_{J,0} + \beta_{J,s(i)} + \beta_{J,g} g_i + \psi_{h(i)}\end{aligned}\tag{6}$$

where g_i is the campaign goal, and $\psi_{h(i)} \sim N(0, \sigma_\psi^2)$ is a hospital random effect on the treatment lift that absorbs differences in surgical quality and post-operative care. The mean-reversion speed is parameterized as $\log(\kappa_i) = \mu_\kappa + \beta_{\kappa T} \cdot z_{T_i}$, where z_{T_i} is a standardized function of the treatment time. This reduced-form specification allows recovery dynamics to vary with the timing of intervention, capturing the clinically relevant idea that earlier treatment may lead to faster recovery by preventing further health deterioration. Under this parameterization, a negative $\beta_{\kappa T}$ implies that earlier-treated patients converge more rapidly toward their post-treatment equilibrium.

All heterogeneity variables entering the equilibrium specifications—clinical pathway, patient age, campaign goal, and hospital—are observed at campaign intake, before any marketing allocation occurs. This ensures that the resulting mapping from treatment timing to health is operational and can be used for ex-ante resource allocation.

The relevant patient-side health outcome object is the expected area under the latent health trajectory over a time horizon H^4 :

⁴ We set $H=365$ days as a common one-year evaluation horizon.

$$E[AUC_i | T_i] = \int_0^H \mathbb{E}[h_i(t) | T_i] dt. \quad (7)$$

Differentiating with respect to treatment timing yields:

$$\frac{\partial AUC_i}{\partial T_i} = -J_i + \Delta_{rev,i}, \quad (8)$$

where the first term reflects the first-order loss from delaying access to the higher post-treatment health level. The second term, $\Delta_{rev,i}$, captures the induced change in recovery dynamics when treatment timing affects the mean-reversion speed.

3.5 Health Return on Marketing

Combining the three links yields the campaign-level Health Return on Marketing, the marginal improvement in patient health per unit of marketing spend, integrating the complete chain from spend to donations, donations to treatment timing, and treatment timing to health. For campaign i , the HROM decomposes as:

$$HROM_{it} = \underbrace{\beta_{S_1,i} \cdot \frac{D_{it}}{S_{it}}}_{\text{Donation response } \left(\frac{\partial D}{\partial S}\right)} \times \underbrace{\beta_{F,i} \cdot \frac{T_i}{G_i}}_{\text{Timing effect } \left(\frac{\partial T}{\partial D}\right)} \times \underbrace{(-J_i + \Delta_{rev,i})}_{\text{Health impact } \left(\frac{\partial AUC}{\partial T}\right)}. \quad (9)$$

The first term converts the elasticity $\beta_{S_1,i}$ into a level derivative, i.e., the additional MU of donations generated by a marginal MU of spend for campaign i . Campaigns with more fundraising traction (high D/S) deliver larger marginal donation returns at the same elasticity; campaigns in segments with higher $\beta_{S_1,i}$ deliver larger returns at the same traction. The second term translates donation response into a change in treatment timing. The AFT model specifies $\log(T_i)$ as linear in funding progress, so $\partial T_i / \partial D_i = \beta_{F_i} \cdot T_i / G_i$, where β_{F_i} is the campaign i 's slope from Equation (3) and G_i is the fundraising goal. The third term $(-J_i + \Delta_{rev,i})$ is the health loss from a one-day delay in treatment.

HROM thus provides a campaign-level measure of the health return on marketing spend. We next characterize the platform's daily marketing budget allocation decision across portfolio of active campaigns, and how HROM can be used for alternative allocation policies that improve patient health.

3.6 Budget Allocation and the Donation–Health Frontier

Each day t , the platform deploys a fixed marketing budget B_t across the campaigns active in their early fundraising window, choosing non-negative spend levels s_i that exhaust the budget. From Link 1, donations respond to spend through the campaign's segment elasticity, which we write in level form as

$$D_i(S_i) = A_i(1 + S_i)^{\beta_i} \quad (10)$$

where s_i is the spend allocated to campaign i , $\beta_i \in (0,1)$ is its segment spend elasticity (in Equation 2), and A_i is its baseline donation productivity, that is, the donations expected at zero marketing spend, given the campaign's intake characteristics and its beginning-of-day state. Because $\beta_i < 1$, D_i is strictly increasing and concave in spend. The productivity term A_i is a model prediction, so the allocation operates on expected donations and is re-solved each day as the state updates; day-1 spend is set by the platform's operational activation and taken as given.

Two benchmarks bound the platform's trade-off. The first ignores health and maximizes total donations,

$$\max_{s_i \geq 0} \sum_i D_i(s_i) \text{ subject to } \sum_i s_i = B_t \quad (11)$$

Since each D_i is strictly concave, the objective is concave and the program has a unique solution.

Equating every funded campaign's marginal donation return to the budget's shadow price,

$\beta_i A_i (1 + s_i)^{\beta_i - 1} = \lambda_t$, yields the closed form

$$s_i^{DM} = \max \left\{ \left(\frac{\beta_i A_i}{\lambda_t} \right)^{\frac{1}{1-\beta_i}} - 1, 0 \right\} \quad (12)$$

with λ_t chosen so that the budget binds; marginal donation returns are thereby equalized across all funded campaigns. This is the donation-maximizing benchmark s_i^{DM} .

The second benchmark ignores donations as an end in themselves and maximizes total patient health, carrying the chain to its conclusion. Let $F_i(s_i) = \frac{D_i(s_i)}{G_i}$ be funding progress and let $\Omega_i(F) = E[AUC_i | F]$ be the expected area under campaign i 's latent health trajectory over the horizon. Writing campaign health as a function of spend, $W_i(s_i) = \Omega_i(F_i(s_i))$, the health-maximizing allocation solves⁵

$$\max_{s_i \geq 0} \sum_i \Omega_i(F_i(s_i)) \text{ subject to } \sum_i s_i = B_t. \quad (13)$$

Because Ω_i is increasing and concave and F_i is concave, the composition W_i is concave in spend, so this objective function also has a unique solution. Equating each funded campaign's marginal health return to a shadow price μ_t ,

$$HROM_i(s_i) = \Omega'_i(F_i) \cdot \frac{1}{G_i} \cdot \frac{\beta_i D_i(s_i)}{1 + s_i} = \mu_t, \quad (14)$$

⁵The theoretical foundations chapter of Neumann et al. (2016) posits that alternative microeconomic approaches are variants of the general principle of "constrained optimization," in particular, the maximization of health or the maximization of individual well-being subject to a budget constraint.

the campaign-level Health Return on Marketing of Equation (9), evaluated at the chosen spend, giving a water-filling solution s_i^{HM} . Unlike donation maximization, it admits no closed form: the health slope $\Omega'_i(F_i)$ is obtained numerically from the joint AFT–OU model, and F_i is nonlinear in spend. We therefore solve it by a one-dimensional search on μ_t ; concavity guarantees that the solution is unique.

These two benchmarks are the endpoints of the donation–health frontier. Interior points maximize health subject to a floor on donations. Indexing policies by a daily donation-retention level $r \in (0,1]$, and writing the donation-maximizing total as $D^{DM} = \sum_i D_i(s_i^{DM})$, the retention- r allocation solves:

$$\max_{s_i \geq 0} \sum_i \Omega_i(F_i(s_i)) \text{ subject to } \sum_i s_i = B_t, \sum_i D_i(s_i) \geq r D^{DM} \quad (15)$$

As r decreases from 1, the floor relaxes and the optimum moves continuously from the donation-maximizing allocation ($r = 1$) toward the health-maximizing one ($r \rightarrow 0$), tracing the frontier; concavity of the health objective makes the frontier itself concave, so the health gained per MU of donation forgone is largest at first and diminishes as more donations are sacrificed. Section 6 computes this frontier day by day on the campaign portfolio, re-solving each program at every campaign’s beginning-of-day state, and reports the resulting retention policies and their operational consequences.

4. SETTING AND DATA

4.1 Institutional Setting

In India, out-of-pocket healthcare expenditure remains high while insurance coverage is limited (WHO, 2024), exposing households to substantial financial risk when major medical

procedures are required (Van Doorslaer et al. 2006, 2007). As a result, households increasingly rely on donation-based crowdfunding to finance care.

Our empirical setting is one of India's largest medical crowdfunding platforms. The unit of analysis is a patient campaign created to finance a specific medical procedure. The platform operates a verified model in which campaigns must provide medical documentation, and funds are disbursed directly to the treating hospital or caregiver. Donations are therefore earmarked to the campaign that attracts them rather than pooled across patients, so that each patient's access to treatment depends on the funds raised by that campaign.

Earmarking has two key implications. First, it makes campaign-level fundraising directly consequential for patient outcomes. Second, it creates a sharp institutional link between funding and access to care: hospitals do not schedule procedures until a campaign has secured sufficient funds. This threshold feature creates a distinctive change in the relationship between fundraising progress and treatment timing once the campaign goal is reached, which we later exploit for identification.

We focus on hospital-managed campaigns, for which the platform controls campaign creation, content, and the allocation of paid digital advertising across channels. This setting allows joint observation of marketing allocation, fundraising, treatment timing, and patient health outcomes within a unified environment. Because the platform determines when and how intensively each campaign is promoted, marketing inputs are observed at the campaign-day-channel level, making exposure allocation directly observable.

The platform retains approximately 15% of each donation as a fee but bears the full cost of paid advertising. As a result, marketing expenditures can exceed commission revenue for individual campaigns, implying that the platform effectively cross-subsidizes campaigns within its portfolio. This creates a clear allocation trade-off: resources are directed toward campaigns where

spending converts into donations most efficiently, even when those campaigns are not where funding most improves patient health.

These institutional features—earmarked donations, threshold-based access to treatment, and centralized control over marketing allocation—jointly generate the variation we exploit to identify the effects of marketing on fundraising, treatment timing, and patient outcomes. More broadly, the setting provides a rare environment in which intermediate marketing decisions and downstream health outcomes are jointly observable.

4.2 Sample Construction and Data Sources

Between 2020 and 2024, the platform hosted 104,573 medical campaigns. Marketing activity and fundraising are highly concentrated: hospital-managed campaigns account for only 2.3% of all campaigns but nearly half of total donations. The analysis spans 1,812 days from January 2020 through December 2024.

We focus on hospital-managed campaigns because the platform directly controls marketing allocation and maintains detailed operational records. Of the 2,382 campaigns in this category, 2,225 posted at least one patient update. After excluding non-informative entries and campaigns with incomplete clinical documentation, the sample reduces to 1,860 campaigns with usable treatment and health information. Within these, we focus on the 1,280 campaigns involving major invasive interventions with well-defined treatment milestones. These procedures account for most fundraising volume and allow reliable measurement of treatment timing. They constitute our estimation sample, and all descriptive statistics below refer to them unless otherwise noted.

We combine four data sources: (1) donation ledgers recording time-stamped contributions, (2) campaign updates containing narrative reports on treatment and recovery, (3) campaign- and patient-level characteristics, and (4) marketing logs capturing daily campaign-level spending by

channel. All data are observed at daily frequency, allowing us to track marketing exposure, fundraising, and treatment progression over time. These measures map directly to the framework: marketing captures allocation, donations capture liquidity, inferred treatment timing reflects access, and health scores measure outcomes. Throughout, the *early window* is defined as the first 14 days following a campaign's first donation.

Table 1 summarizes the 1,280 campaigns in the analysis sample. Campaigns exhibit substantial heterogeneity in both marketing exposure and fundraising outcomes. Early-window paid marketing averages 6,900 MU (median 1,700 MU), while early-window donations average 16,800 MU (median 3,500 MU), indicating a highly skewed distribution in which a relatively small subset of campaigns accounts for a large share of both spend and contributions. Paid spend is distributed across three channels—Meta (mean 41% of campaign spend), Google/YouTube (36%), and display (23%)—a composition that becomes central to the identification strategy. The platform's Social Impact matching program contributes an average of 2,600 MU but is zero for the median campaign, indicating that matching funds are concentrated among a subset of cases.

Table 1: Campaign and marketing characteristics

Variable	Mean	SD	Median
Goal	20.22	13.72	18.29
Marketing window (days)	59.64	94.75	25.00
Activation delay (days)	9.32	15.94	6.00
Marketing spend (early)	6.87	13.90	1.69
Donations (early)	16.84	35.01	3.49
Marketing spend (full campaign)	11.75	22.32	4.19
Donations (full campaign)	31.20	50.07	12.75
Platform matching	2.63	11.86	0.00
Meta share of spend	41.10	37.22	35.41
Google share of spend	35.55	36.12	24.08
Display share of spend (%)	23.35	33.71	5.93

Notes: Campaign-level summary statistics for the 1,280-campaign analysis sample. Monetary variables are in thousands of monetary units (MU). The early window is the first 14 days from a campaign's first donation; the marketing window is the span from first to last observed paid spend. Activation delay is the number of days from the first donation to the first paid marketing spend and is defined for the 1,280 campaigns that receive any paid marketing.

Channel shares are computed over full-campaign spend. Platform matching denotes funds contributed by the platform's Social Impact program. Funding-progress and patient measures are reported in Table 3.

Marketing activity is heavily front-loaded: the first two weeks account for more than half of both total donations and paid marketing expenditures. Because early funding governs access to treatment, this concentration makes initial allocation decisions disproportionately consequential for patient outcomes. Marketing expenditures are also large relative to early donations, so marginal reallocation can meaningfully shift funding trajectories.

4.3 Platform Marketing Operations

Three features of the allocation environment generate identifying variation: (i) concentration of daily spending across campaigns, (ii) quasi-random timing of campaign activation, and (iii) exogenous variation in channel availability.

Budget concentration and allocation environment. Marketing is concentrated in the early window (the first 14 days from a campaign's first donation). We focus on this period because it is when paid promotion has the greatest influence on fundraising and, in turn, treatment timing.

Table 2: Platform allocation environment

Variable	N	Mean	SD	Median
Total daily budget	1,717	8.33	7.38	6.18
Campaigns with positive spend	1,717	3.72	2.02	3.00
Outside option share	1,717	43.66	32.99	36.53
Top 1 share within inside	1,717	75.09	20.27	77.38
Top 3 share within inside	1,717	97.46	5.07	100.00

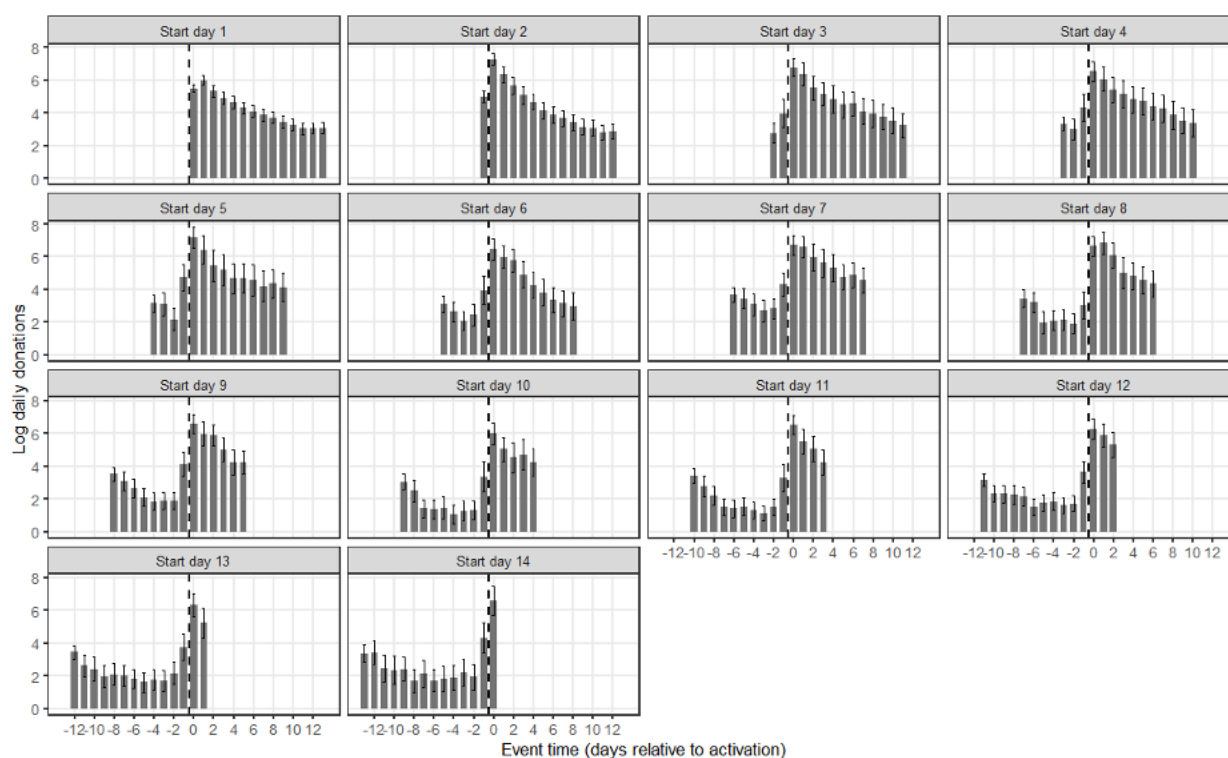
Notes: Daily-level summary statistics for the platform's allocation environment. The unit of observation is a platform-day, restricted to days with at least one inside (early-window) campaign receiving positive marketing spend. Total daily budget is in thousands of monetary units. "Campaigns with positive spend" is the number of campaigns receiving strictly positive paid marketing on a given day. "Outside option share" is the percentage of the daily budget allocated to campaigns outside their early window. Top-k shares are computed within the inside portfolio and give the percentage of inside spend allocated to the top k campaigns.

Table 2 summarizes the daily allocation environment across 1,717 platform-days, defined as days on which at least one early-window ("inside") campaign receives paid marketing. On a typical day, the platform deploys a median budget of 6,200 MU (mean 8,300 MU), with substantial

day-to-day variation that creates meaningful fluctuations in total exposure. A sizable share of this budget—44% on average (median 37%)—is directed to campaigns outside their early window, leaving the remainder to be allocated across inside campaigns.

Within the inside portfolio, allocation is highly concentrated. A median of three campaigns receives positive spend on a given day, with the largest recipient capturing 77% of inside spend and the top three accounting for essentially all of it (median 100%, mean 97%). Such concentration implies that small changes in allocation produce large differences in campaign visibility and funding trajectories.

Figure 2: Donations around paid marketing activation by cohort



Notes: N = 1,053 campaigns (227 of the total 1,280 campaigns activate paid marketing outside the 1-14 days early fundraising window). Each panel is a cohort defined by the respective marketing activation day. Daily donation reported in MU.95% confidence bands.

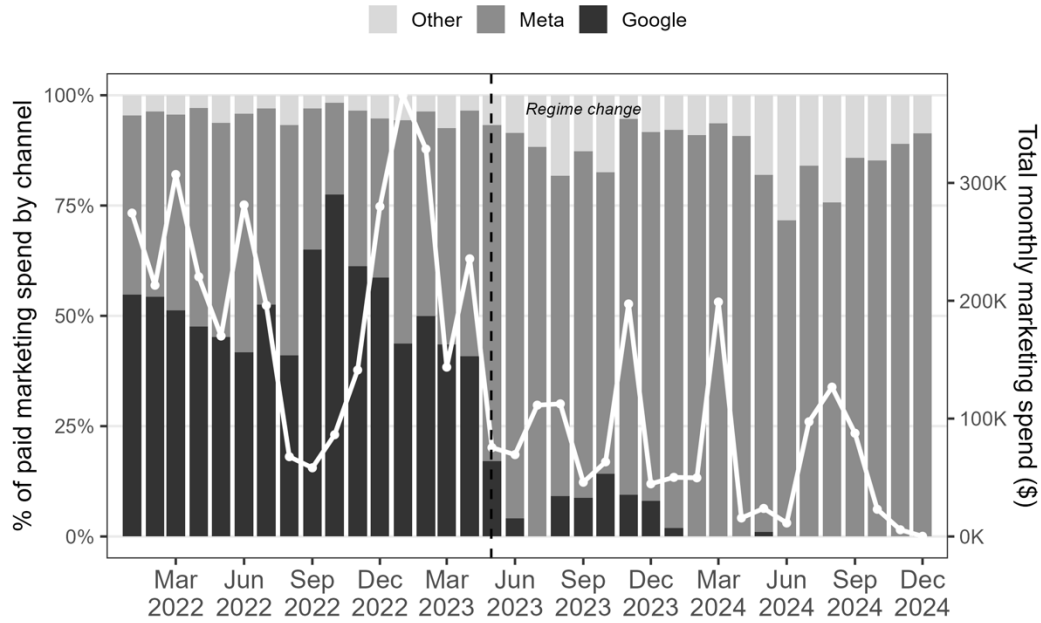
Timing of Marketing Campaigns. Paid marketing does not begin immediately for all campaigns. The median campaign is activated six days after its first donation (Table 1), but activation timing is highly dispersed: some campaigns are activated on the day of the first donation, others only after delays of several days or weeks. Discussions with platform managers indicate that this variation reflects operational constraints—delays in creative production, compliance checks, and patient coordination—rather than expected fundraising performance.

Most campaigns concentrate paid spend in a single early activation event, so the timing of activation provides clean, largely one-time variation in marketing exposure. Figure 2 plots mean daily donations in event time relative to activation, restricting attention to cohorts activating paid marketing within their early window of fundraising. There is a discontinuous change in daily donations on the activation day, and this spike is consistent with a causal effect of marketing exposure on fundraising.

Regulatory channel shock. In July 2023, the Advertising Standards Council of India (ASCI) introduced guidelines restricting advertisements for charitable causes, imposing content requirements on ads featuring individuals in medical distress (ASCI, 2023). These rules limited the use of emotionally charged visuals and required additional disclosure, disproportionately affecting short-form video formats in which visual storytelling is central.

The Google channel—primarily YouTube video advertising—was particularly affected. Before the policy change, this channel was a core vehicle for disseminating patient narratives optimized for engagement. The new guidelines reduced the platform's ability to deploy such creatives, lowering both the attractiveness and feasibility of Google-based campaigns relative to other channels. Because the policy was externally imposed and applied uniformly across campaigns, it generates exogenous variation in the set of available marketing channels.

Figure 3: Platform advertising channel shares and total monthly marketing spend



Notes: Bars show the Google (YouTube), Meta (Facebook/Instagram), and Other (open-web display and remaining channels) shares of total platform paid advertising spend by month (left axis), January 2022 through December 2024. The white line shows total monthly marketing spend in MU (right axis). The dashed line marks the regulatory regime change.

Figure 3 shows a sharp, system-wide reallocation of spend around the policy change. Google's share of paid marketing fell from roughly 40–55% before the regime change to below 10% by mid-2023, approaching zero through 2024, effectively removing one of the platform's primary acquisition channels; over the full sample period Google still accounts for 36% of spend on average (Table 1). Total monthly marketing spend in MU (right axis) decreased, as the shock operated as a contraction of *available channels*, concentrating activity onto fewer of them and reducing available channels for spending. The abrupt, discrete nature of the shift is consistent with an externally imposed constraint on channel feasibility. We exploit this channel removal, combined with the quasi-random activation delays, to identify the effect of marketing spending on donations.

4.4 Clinical Pathways and Patient Health Patterns

We classify campaigns into seven clinical pathway segments using a priority-ordered classifier applied to LLM-coded medical text from campaign narratives: (1) HSCT/BMT for nonmalignant conditions and primary immunodeficiencies (HNI); (2) HSCT/BMT for hematologic malignancies (HHM); (3) Organ transplant (OTx); (4) Cancer non-HSCT (CnH); (5) Cardiac surgery (CSx); (6) Non-cardiac surgery (NSx); and (7) Other or unclassified (Oth). These categories capture systematic differences in clinical severity, treatment requirements, and recovery profiles, and motivate segment-specific parameters in the structural model. For estimation, each pathway is interacted with patient age group (adult, child, infant). Because infant campaigns are sparse in several pathways, infant cells are pooled into four broader categories: HSCT/BMT pathways combined with infant cancer cases, organ transplant, cardiac surgery, and non-cardiac surgery combined with other procedures. This yields eighteen pathway-by-age segments.

Table 3 summarizes the health subsample. The median patient is 8 years old. Early-window fundraising progress varies widely: the median campaign raises 24% of its goal in the first 14 days, with 26% reaching full goal attainment in this window; over the full campaign, the median reaches 94% of goal and 48% of campaigns eventually meet their target.

Table 3: Patient, funding progress, and health outcomes

	N	Mean	SD	Median
Patient age (years)	1,280	10.54	10.20	8.00
% goal raised, early window	1,280	112.27	234.93	23.90
% reaching goal, early window	1,280	26.25	44.02	—
% goal raised, full campaign	1,280	188.95	284.98	94.19
% reaching goal, full campaign	1,280	47.97	49.98	—
Reporting treatment (%)	1,280	62.89	48.33	—
Days to treatment, L_i (treated)	805	80.28	135.53	29.26
Days to treatment, R_i (treated)	805	178.98	192.14	111.51
Days to last update (untreated)	475	347.49	303.25	248.57
Pre-treatment health score (1–5)	1,026	2.34	0.78	2.00

	N	Mean	SD	Median
Post-treatment health score (1–5)	711	3.77	1.05	4.00

Notes: Treatment dates are not directly observed. The two bracket rows (L_i , R_i) bound the interval within which treatment occurred for treated campaigns; the AFT model (Section 3.3) recovers the treatment-date distribution from these bounds. Untreated campaigns are right-censored: treatment may have occurred after the last observed update or not at all. Binary variables report the mean only (median suppressed). Health scores are on a 1–5 scale (higher = better), generated by Claude Sonnet using condition-specific rubrics; pre-treatment scores are available for the 1,026 campaigns with at least one update before L_i , and post-treatment scores for the 711 treated campaigns with at least one update after R_i .

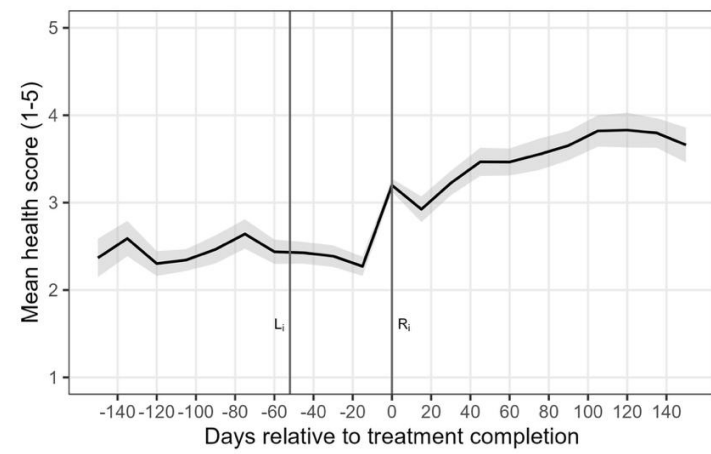
Treatment timing is not directly observed. Campaign updates provide narrative descriptions of patient progress, but the exact date of treatment is never recorded. For the 805 campaigns (63%) that report treatment completion, we observe only that treatment occurs between two update events: after the last update that does not mention completion and by the first update that confirms it. This defines an interval within which treatment must have occurred.

In survival-analysis terms, these observations are interval-censored: the event is known to occur, but its exact timing lies within a window rather than being observed at a single point. The remaining 475 campaigns (37%) never report treatment during the observation period; for these, the last observed update provides a lower bound on treatment timing, making them right-censored observations.

Patient Health Measurement: We use Claude Sonnet to convert each timestamped patient update text into a composite timestamped health score on a 1–5 scale, with higher values indicating better health. The scoring rubric synthesizes information across dimensions aligned with the EQ-5D framework (EuroQol Group 1990), a generic health status measure designed to capture health across diverse clinical conditions and settings (Devlin and Brooks 2017). The dimensions include mobility, self-care and care dependency, usual activities and functional status, pain and discomfort, and emotional well-being. We validate the scoring protocol through an independent rescoring exercise using Claude Haiku that yields near-perfect agreement. The LLM prompt and procedure

used for scoring along with results from the validation exercise are reported in Online Appendix. *Model-Free Evidence in Patient Health Patterns.* First, our data show that treatment produces a discrete and persistent improvement in health. Figure 4 plots mean health scores aligned in event time relative to the first update reporting treatment completion (R_i). Health is relatively flat beforehand, fluctuating between roughly 2.3 and 2.6 and dipping to about 2.25 in the weeks just before treatment, then rises sharply at completion to approximately 3.2. It continues climbing over the following months to a long-run plateau near 3.7–3.8 before easing slightly. This pattern—a discrete jump followed by mean-reverting progress—motivates modeling health as a continuous-time process with a treatment-induced shift in the long-run equilibrium.

Figure 4: Patient health trajectory around treatment milestone

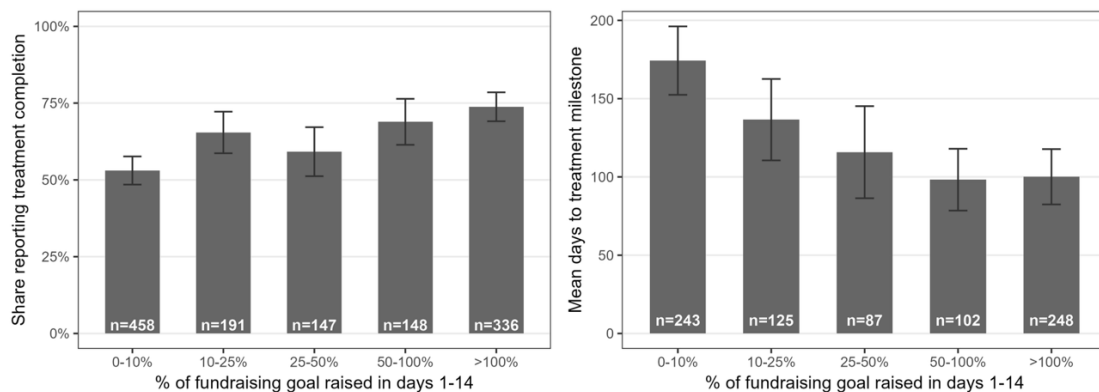


Notes: $N = 805$ treated campaigns. Mean health score (1–5 scale) in 15-day bins relative to treatment confirmation (R_i). Shaded band: 95% confidence interval. Vertical lines mark median L_i and R_i .

Second, early fundraising affects both access to treatment and its timing. Figure 5 documents two margins of this relationship. On the extensive margin, campaigns that raise a larger share of their goal in the first 14 days are more likely to report treatment completion within the observation window: the share rises from 57% in the lowest funding bin (0–10% of goal) to 78% in the highest (>150% of goal), though intermediate bins exhibit some noise. On the intensive margin, conditional on treatment being observed, early funding substantially accelerates care.

Measured at the midpoint of the treatment interval $[L_i, R_i]$ reported in Table 3, mean time to treatment declines from 175 days for campaigns raising less than 10% of goal to 81 days for those exceeding 150%—a reduction of more than three months. This pattern reflects the institutional mechanism: hospitals condition treatment scheduling on demonstrated funding, so earlier liquidity translates directly into faster access, which is then collected during platform-initiated follow-up calls or in some cases reported voluntarily by the patients.⁶

Figure 5: Treatment probability and treatment timing by early fundraising progress



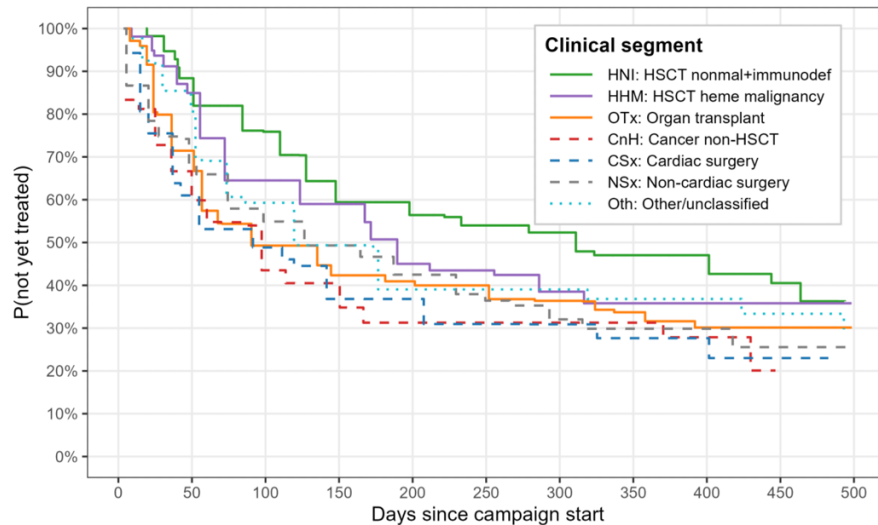
Notes: Left: share reporting treatment completion. Right: mean days to treatment among 805 treated campaigns. Error bars show 95% confidence intervals. n values show campaigns per bin.

Because treatment is observed only within a finite window, campaigns that never report treatment are right-censored: treatment may occur after the sample period but is unobserved. The duration model does not separate delayed treatment from treatment that never occurs; it treats every unobserved case as occurring beyond the window. Early fundraising therefore shapes not just *when* treatment happens but *whether* it is observed at all—slower fundraising lengthens delays and pushes treatment past the censoring point, while faster fundraising pulls it into view. The

⁶ Campaign narratives frequently state that treatment is delayed explicitly due to insufficient funds, reinforcing the interpretation that fundraising progress is a binding constraint on access to care. Illustrative excerpts from updates include “We are failing as parents to save our child’s life as we still haven’t managed to raise the required funds for her treatment. Her heart transplant is still pending,” and “Second surgery getting delayed due to shortage of funds. If her condition is not corrected, our daughter will suffer her whole life.”

gradient in observed completion rates should thus be read partly as a timing-and-censoring effect rather than a pure difference in the probability of ever being treated. At the same time, some of this acceleration likely reflects genuine gains in access for patients who would otherwise face very long delays or never receive treatment at all.

Figure 6: Time to treatment by clinical pathway segment



Notes: Curves show the estimated probability of not yet having received treatment by day t since campaign start, computed using the Turnbull nonparametric maximum likelihood estimator (NPMLE) to account for interval-censored treatment timing. $N = 1,280$ campaigns.

Third, treatment timing varies substantially across clinical pathways. Figure 6 plots the probability of not yet having received treatment over time for each segment. The two HSCT pathways (HNI and HHM) progress most slowly: their untreated probability stays above 50% past roughly 180 days and remains elevated—near 36%—even at 500 days. These long horizons reflect clinical constraints such as donor-compatibility screening and transplant preparation, which funding alone cannot accelerate. Cancer non-HSCT (CnH) and cardiac surgery (CSx) lie at the other extreme, declining most steeply: their untreated probability falls below 30% within about five months and reaches roughly 20–23% by the end of the window. For these procedures, timing is governed more directly by funding availability than by clinical scheduling, so early liquidity translates more readily into faster treatment. The remaining segments fall between these poles.

Together, these descriptive patterns characterize the variation in the data that allows us to estimate segment-specific parameters in the M2H model. We turn to estimation and identification in the next section.

5. ESTIMATION AND IDENTIFICATION

This section describes the estimation strategy for each component of the sequential M2H framework and the identification arguments for the estimated parameters.

We start by noting that we randomly split the full sample of 1,280 campaigns into an estimation sample of 1,017 campaigns and a holdout sample of 263 campaigns. All parameters are estimated on the 1,017-campaign estimation sample; the holdout sample is reserved for out-of-sample evaluation: the misalignment diagnostics below are computed on the holdout campaigns, while the counterfactual simulations apply the estimated parameters to the full campaign portfolio. The random splitting ensures both samples contain campaigns from both the pre- and post-Google-shutdown regimes, avoiding confounds between the regime change and the out-of-sample evaluation.

5.1 Donation Model

The first link, the donation-spend model, is estimated by Poisson pseudo-maximum likelihood (PPML). Although daily donations are a continuous non-negative outcome, PPML is preferred over OLS in logs because it directly estimates the log-linear conditional mean without requiring a log transformation of the dependent variable. The PPML estimator is consistent for the conditional mean parameters for any non-negative dependent variable, regardless of the actual distribution (Gourieroux et al. 1984; Santos Silva and Tenreiro 2006) and is robust to the heteroskedasticity in the spend–donation relationship.

A central empirical challenge in estimating this model is that platform marketing spend is not randomly assigned across campaigns or over time. The platform is likely to allocate advertising based on recent donation performance and other signals of campaign appeal, meaning that campaigns receiving higher spend are also those expected to generate stronger donation responses. To control for this endogeneity, we use a control-function approach. We use two instruments. The first one exploits variation in activation delays. The instrument is:

$$Z_{it} = t_i^{act} \times delay_i \quad (16)$$

where t_i^{act} is the first day campaign i receives paid spend and $delay_i$ is the number of days from campaign launch to first activation. The instrument is nonzero only on the activation day itself, providing a one-time exogenous shock to total spend at the moment marketing first turns on. The second instrument exploits the mid-2023 regulatory restriction on charitable-cause advertising described earlier, which effectively eliminated Google/YouTube as a marketing channel while Meta/Facebook remained available for campaign promotion. This channel shock generated a system-wide reduction in total available spend that is orthogonal to individual campaign donation demand. The activation-delay instrument captures the extensive-margin transition from zero to positive spend, while the Google/YouTube shutdown instrument provides additional intensive-margin variation after activation. Spend is empirically concentrated near the activation date, so the activation-day shock captures the dominant share of within-campaign spend variation for most campaigns. Together, the two instruments isolate exogenous variation in total marketing spend arising from platform operating constraints rather than contemporaneous donor demand, supporting identification of the spend elasticity across the full early window.

The first stage regresses $\ln(1+S_{it})$ on both instruments, the observable campaign characteristics X_i , lagged and cumulative donations, campaign age, and the post-shutdown regime indicator P_t (is_post):

$$\ln(1 + S_{ikt}) = \pi_k \mathbf{Z}_{ikt} + \mathbf{x}'_{it} \gamma_k + \alpha_{ik} + \xi_{ikt}. \quad (17)$$

The first-stage residual $\hat{v}_{it} = \ln(1 + S_{it}) - E[\ln(1 + S_{it})]$ serves as the control-function term in the second-stage PPML.⁷ A significant coefficient indicates that residual endogeneity is present for that segment and that the correction is warranted (Terza et al. 2008). Standard errors are cluster-robust at the campaign level.

5.2 Treatment Timing and Health Trajectory

The AFT model for treatment timing and the OU model for latent health dynamics are estimated jointly in a Bayesian data-augmentation framework (Rossi et al. 2005). Joint estimation is necessary because the latent treatment time $\mathbf{T}_i$ enters both the duration likelihood and the OU transition equations: conditioning on a sampled $\mathbf{T}_i$ determines which OU regime applies at each observation. For censored campaigns, the AFT provides the predictive distribution of treatment times, while the observed health trajectory helps place treatment within the censored interval, sharpening the latent time draws. Data augmentation resolves this interdependency by iterating between drawing latent treatment times and updating model parameters, as described in detail below.

AFT likelihood for interval- and right-censored observations. As previously described, the platform does not observe exact treatment dates. Instead, for each campaign, the true treatment

⁷A basic control function requires only a single first-stage residual in the second stage, which yields consistent segment-specific spend coefficients even though spend is interacted with segment indicators (Ebbes, Papies, and van Heerde 2022). We adopt the more flexible correlated-random-coefficient form of Wooldridge (2015, Eq. 29), which adds the first-stage residual and its interactions with elements of the second-stage regressor vector.

time T_i is known to lie within an interval $(L_i, R_i]$, where L_i is the last update confirming treatment had not yet occurred and R_i is the first update confirming it had. Campaigns that never report treatment within the observation window are right-censored at C_i .

Because the exact time is not observed, the usual density-based likelihood contribution $f(t_i)$ is not available. The correct likelihood contribution is the probability that T_i falls in the observed region. For an interval-censored observation:

$$Pr(T_i \in (L_i, R_i]) = F(R_i) - F(L_i) = (1 - S(R_i)) - (1 - S(L_i)) = S(L_i) - S(R_i),$$

where $F(t) = Pr(T_i \leq t)$ and $S(t) = 1 - F(t) = Pr(T_i > t)$, i.e., the survivor function. For a right-censored observation, the likelihood contribution is the probability that treatment has not yet occurred by the censoring date, $Pr(T_i > C_i) = S(C_i)$

With log-normal AFT errors, the survivor function takes the form:

$$S(t | \cdot) = 1 - \Phi\left(\frac{\log(t) - \eta_i}{\sigma}\right), \quad (18)$$

where η_i collects the linear predictor from Equation 4 and Φ is the standard normal CDF.

Substituting the two cases above, the individual likelihood contribution is:

$$\mathcal{L}_i^{\text{AFT}} = \begin{cases} S(L_i | \cdot) - S(R_i | \cdot), & \text{interval-censored } (T_i \in (L_i, R_i]) \\ S(C_i | \cdot), & \text{right-censored } (T_i > C_i). \end{cases} \quad (19)$$

In both cases, a larger value of η_i —corresponding to slower expected treatment from lower early fundraising—shifts the log-normal distribution to the right, reducing $S(L_i) - S(R_i)$ and thus the predicted probability that treatment falls in the observed window. This is the mechanism through which the AFT model links early fundraising to both the timing and the observed completion of treatment.

The key identifying assumption in Equation (3) is that, conditional on clinical pathway, hospital, campaign goal, baseline health, and observed fundraising states, early funding progress shifts treatment timing but is not otherwise correlated with unobserved determinants of treatment

timing. The institutional details discussed earlier support this interpretation Campaign presentation and promotion are platform-managed; earmarking and hospital financing thresholds link fundraising progress directly to access to care. The spline with a knot at $F_{it} = 1$, uses the threshold logic of the campaign goal, capturing the change in the relationship between fundraising progress and treatment timing once sufficient funds have been raised, which helps address reverse-causality concerns and strengthen identification.

In terms of heterogeneity, the hierarchical prior on the segment-specific below-goal slopes shrinks estimates for segments with fewer observations toward the pooled mean while allowing well-identified segments to depart from it; hospital random intercepts $\phi_{h(i)} \sim N(0, \sigma_\phi^2)$ absorb persistent differences in scheduling capacity across hospitals.

Conditional on the sampled treatment time T_i , the OU process implies a closed-form Gaussian transition between consecutive observations (Uhlenbeck and Ornstein, 1930). Between update times $t_{i,j-1}$ and t_{ij} with gap $\Delta_{ij} = t_{ij} - t_{i,j-1}$, the latent health evolves as

$$h_i(t_{ij}) = \mu_i(t_{ij}) + e^{-\kappa_i \Delta_{ij}} [h_i(t_{i,j-1}) - \mu_i(t_{i,j-1})] + \varepsilon_{ij}^{\text{proc}}, \quad (20)$$

where $\mu_i(t)$ is the regime-dependent equilibrium from Equation (6), $\kappa_i > 0$ is the campaign-specific mean-reversion speed, and $\varepsilon_{ij}^{\text{proc}} \sim N(0, \sigma_X^2(\Delta_{ij}))$ with $\sigma_X^2(\Delta_{ij}) = \sigma_X^2 \cdot (1 - e^{-2\kappa_i \Delta_{ij}}) / (2\kappa_i)$. Short gaps allow little latent drift; long gaps allow more variability to accumulate. Here σ_X governs the scale of latent health fluctuations between updates. Each observed health score is a noisy snapshot of this latent state: $y_{ij} | h_i(t_{ij}) \sim N(h_i(t_{ij}), \sigma_Y^2)$, where σ_Y is the measurement-noise standard deviation capturing update coding noise. The irregular spacing of updates helps identify σ_X from σ_Y : process variance accumulates with gap length while measurement noise does not (Durbin and Koopman, 2012).

The posterior distribution is proportional to the product of the measurement density, the OU transition density, and the interval/right-censored AFT likelihood:

$$p(\theta, \mathcal{X}, \mathcal{T} \mid Y) \propto \prod_{i,j} p(y_{ij} \mid h_i(t_{ij}), \sigma_Y) \times \prod_{i,j} p(h_i(t_{ij}) \mid h_i(t_{i,j-1}), T_i, \theta^{\text{OU}}) \times \prod_i \mathcal{L}_i^{\text{AFT}}(\theta^{\text{AFT}}; L_i, R_i) \times p(\theta^{\text{OU}}) p(\theta^{\text{AFT}}), \quad (21)$$

where $\mathcal{X} = \{h_i(t_{ij})\}$ collects the latent health states and $\mathcal{T} = \{T_i\}$ the latent treatment times.

Each MCMC iteration cycles through three blocks: (i) latent health states $\mathcal{X}$, drawn via a forward-filter/backward-sample pass that exploits the Gaussian OU transition (Durbin and Koopman, 2012); (ii) latent treatment times $\mathcal{T}$, each drawn from its log-normal full conditional truncated to $(L_i, R_i]$ for interval-censored observations or (L_i, ∞) for right-censored; and (iii) model parameters θ^{AFT} and θ^{OU} , updated via Gibbs steps for conjugate components and adaptive Metropolis for non-conjugate terms. We specify diffuse Normal priors for regression coefficients and Gamma priors for precision parameters; specific prior choices are reported in the Online Appendix.⁸ As in the AFT, segment-level OU parameters are estimated with hierarchical shrinkage, regularizing estimates for segments with few campaigns.

In terms of identification, Table 4 highlights the variation in the data that allows us to identify the parameters for both the marketing-donation model and the patient-side model (treatment timing and health trajectory).

Table 4: Parameter blocks and identifying variation

Parameters	What it governs	Identifying variation
Spend elasticity ($\beta_{SL,i}$)	Causal donation lift from marketing spend, heterogeneous by clinical-pathway $\times$ age segment	Activation-delay instrument (Z_{it}) and the regulation-driven 2023 Google/YouTube ads shutdown (post-shutdown indicator P_t); two sources of exogenous variation in total spend,

⁸We run two parallel chains with 5,000 adaptation iterations, 5,000 burn-in, and 20,000 sampling iterations. Convergence is assessed via the Gelman–Rubin $\hat{R}$ statistic (Gelman and Rubin, 1992) and effective sample sizes; all reported parameters satisfy $\hat{R} < 1.05$ with effective sample sizes exceeding 500. Posterior medians serve as point estimates and 95% posterior credible intervals provide inference.

Parameters	What it governs	Identifying variation
		isolated via the control-function residual $\hat{v}_{it}$ (2SRI)
Donation momentum and saturation (γ_{D1}, γ_{D2})	Day-to-day persistence in donations; declining returns as the donor pool is tapped	Within-campaign serial correlation in donations (γ_{D1}); declining donation trend as cumulative donations accumulate (γ_{D2})
Cumulative spend saturation (β_{S2})	Diminishing returns from repeated marketing exposure within a campaign	Within-campaign variation in cumulative past spend, net of current-period spend and the control-function residual
Campaign-level controls ($\alpha'X_i$)	Baseline donation level unrelated to marketing	Pre-activation and day-0 donations, together with hospital, pathway, and age-group fixed effects, isolate organic donation levels net of marketing
Funding $\rightarrow$ timing slopes (β_{F1i}, β_{F2i})	How fundraising accelerates treatment, below vs. above the goal; varies by clinical pathway	Cross-campaign variation in early funding conditional on segment, goal, and hospital; hierarchical shrinkage for small segments
Goal and matching effects (β_G, β_M)	Financial barrier from procedure cost; platform co-investment	Cross-campaign variation in goal size and matching conditional on segment and hospital
Hospital random effects ($\varphi_{h(i)}$)	Scheduling-capacity differences across hospitals	Within-hospital variation in treatment timing net of funding and segment
Pre-treatment health ($\beta_{\mu,0}, \beta_{\mu,s(i)}, \beta_{\mu,g}$)	Baseline health level; segment and goal-size heterogeneity	Pre-treatment health observations within and across segments and goal sizes
$J_i: \beta_{J,0}, \beta_{J,s(i)}, \beta_{J,g}, \psi_{h(i)}$	Discrete health improvement at treatment; hospital heterogeneity in surgical/care quality	Discontinuity in health scores around treatment reports; segment- and goal-size variation in jump magnitude; within-hospital variation in lift magnitude net of segment and goal
$\mu_\kappa, \beta_{\kappa T}$	Post-treatment mean-reversion (recovery) rate, and its dependence on treatment timing	Decay of serial dependence across irregularly spaced post-treatment updates (identifies μ_κ); covariation between treatment timing and the post-treatment convergence rate (identifies $\beta_{\kappa T}$)
Process vs. measurement noise (σ_X, σ_Y)	Latent health drift vs. measurement noise in update reports	Gap-length scaling: variance growing with Δ_{ij} (identifies σ_X); residual variance independent of spacing (identifies σ_Y)

Notes: Parameters correspond to the donation model (Equation 2) and its control-function first stage (Equations 16–17), the AFT treatment-timing model (Equations 3–4), and the OU health-trajectory model (Equations 5–6).

6. RESULTS

We now present the results in three parts. First, we report the parameter estimates for the donation model (Link 1), focusing on the segment-level spend elasticities and intercepts that drive

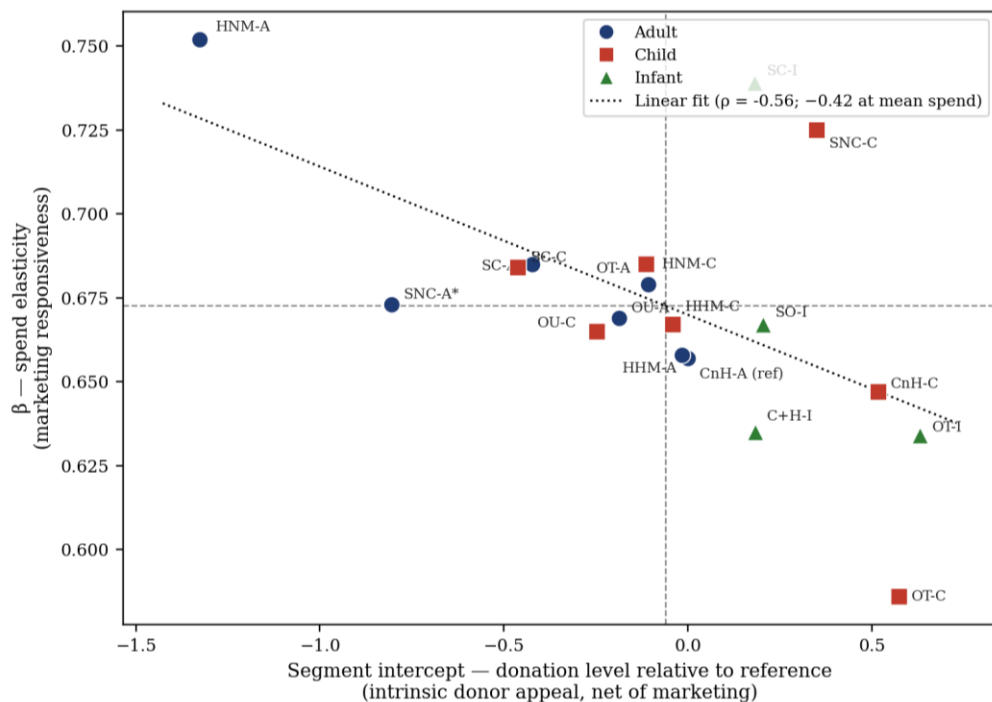
heterogeneity in donation response. Second, we report the joint AFT–OU model’s parameter estimates (Links 2–3). Third, we use the estimated parameters to simulate the counterfactual budget allocation and trace the donation–health frontier. .

Donor Appeal Across Clinical Pathways

In Figure 7, we plot two sets of segment-level estimates from the donation model: the segment-specific spend elasticity ($\beta_{S_1,i}$), which measures the percentage increase in donations from a one-percent increase in marketing spend, against the segment intercept, the donation level each segment attracts regardless of marketing. (e.g., Cancer non-HSCT, Adult), which measures intrinsic donor appeal.⁹ Under the log-linear specification, both objects enter the marginal donation return: the return in levels is proportional to $\beta_{S_1,i} \frac{D_{it}}{S_{it}}$, where D_{it} inherits $\exp(\alpha_i)$ through the multiplicative structure.. How these two components relate to each other therefore determines where a donation-maximizing allocation would direct marketing spend.

Figure 7: Two Dimensions of Donor Appeal Across Clinical Pathways

⁹ The first stage results estimating the control function residuals and the estimates of the remaining variables in the second stage are described in the Online Appendix. We also include some additional measure of model fit.



HNH = HSCT/BMT for nonmalignant conditions and primary immunodeficiencies; HHM = HSCT/BMT for hematologic malignancies; OT = Organ Transplant; CnH = Cancer non-HSCT; SC = Cardiac Surgery; SNC= Non-Cardiac Surgery; OU = Other/unclassified. Suffix denotes age group (A = Adult, C = Child, I = Infant). C+H-I pools sparse infant cells HNM+HHM+CnH; SO-I pools infant cells SNC+OU.

The segment-specific spend elasticities range from 0.586 (Organ Transplant, Children) to 0.752 (HSCT non-malignant + Immunodeficiency, Adults). The elasticities are economically meaningful (statistically significant at $p < 0.01$) and every estimate lies below one, confirming diminishing marginal returns to marketing spend across all clinical pathways.

The two dimensions in the figure are negatively correlated across pathways ($\rho = -0.56$; this correlation drops to -0.42 but still significant when the donation baseline, i.e., the appeal of the segment, is evaluated at the mean campaign spend rather than at zero spend, confirming the relationship is not an artifact of zero-spend extrapolation). However, the variation in intercepts across segments is roughly an order of magnitude larger than the variation in log-elasticities. Consequently, the marginal donation return in levels remains highest for high-appeal segments throughout the observed spend range: the intercept advantage more than compensates for the elasticity disadvantage. A donation-maximizing allocator would therefore rationally concentrate

spend on campaigns with high intrinsic appeal campaigns — the anticorrelation notwithstanding.

We find that age is a significant factor of campaign appeal. Children and infants carry higher donor appeal than adults (average segment intercept of +0.08 for children and +0.30 for infants, versus −0.41 for adults), consistent with greater emotional salience of campaigns for younger patients. The average spend elasticity is similar across the three age groups, suggesting that the marginal MU of marketing is similarly productive for all segments, even the less appealing to donors. Donation-maximizing allocation would therefore concentrate spend on infant and child campaigns. Whether this concentration serves patient health or whether the health return per marketing dollar follows a different pattern is the question the remaining links of the M2H chain address..

Treatment Timing and Health Trajectory

The second link of the M2H chain examines how early fundraising progress affects treatment timing. Table 5 reports posterior estimates from the joint AFT–OU model, estimated by Bayesian MCMC on the 1,017-campaign estimation sample.

The global intercept ($\alpha_0 = 5.117$) implies a baseline median time to treatment of approximately 167 days, in the absence of any early fundraising progress. Early fundraising progress below the campaign goal significantly accelerates treatment ($\mu_1 = -0.757$), with meaningful heterogeneity across clinical pathways ($\sigma_1 = 0.575$). Fundraising beyond the goal has a smaller additional effect ($\beta_{F_2} = -0.303$). This asymmetry is consistent with the institutional setting: most of the treatment-timing benefit comes from crossing the financial barrier required to schedule surgery, with limited additional acceleration once the threshold is met. Campaigns with larger goals face longer treatment delays ($\beta_g = 0.647$), reflecting the greater financial barrier to

more complex procedures. The platform’s matching contribution does not significantly impact treatment timing.

Table 5: Joint AFT–OU Estimates

Parameter	Estimate	Post. SD	95% CrI
A. Treatment timing (AFT)			
Intercept	5.117*	0.143	[4.834, 5.408]
Below-goal funding progress, mean	−0.757*	0.219	[−1.199, −0.343]
Below-goal funding progress, across-segment SD	0.575*	0.132	[0.384, 0.893]
Beyond-goal funding progress	−0.303	0.155	[−0.598, 0.012]
Campaign goal size	0.647*	0.182	[0.296, 1.016]
Platform matching funds	0.015	0.072	[−0.131, 0.152]
Treatment-time scale, log-normal	1.761*	0.075	[1.623, 1.914]
SD of segment AFT intercepts	0.424*	0.081	[0.308, 0.624]
B. Health dynamics (OU)			
Pre-treat. Health – mean	2.253*	0.075	[2.099, 2.396]
Pre-treatment health — goal effect	−0.258*	0.043	[−0.353, −0.187]
Health gain at treatment — mean	1.224*	0.176	[0.843, 1.539]
Health gain at treatment — goal effect	−0.589*	0.114	[−0.815, −0.372]
Recovery speed	−4.957*	0.060	[−5.082, −4.843]
Earlier treatment → faster recovery	−0.569*	0.078	[−0.725, −0.421]
Latent-health SD	0.099*	0.002	[0.094, 0.104]
Observation SD	0.613*	0.008	[0.599, 0.629]
SD of hospital effect on treatment lift	0.46	0.059	[0.362, 0.587]
Sample			
Campaigns		1,017	
Health observations		7,270	

Notes: Bayesian estimation via MCMC; posterior means with posterior SD and 95% credible intervals. * denotes a 95% credible interval excluding zero. Estimation sample: 1,017 campaigns (excludes 263 holdout campaigns).

The third link examines how treatment and its timing shape patient health trajectories. Pre-treatment health averages 2.253 on the 1–5 scale for the reference segment, with sicker patients (larger goals) starting lower ($\beta_\mu, g = -0.258$). Treatment produces a discrete health improvement ($\beta_J, 0 = 1.224$), with the gain diminishing for more severe cases ($\beta_J, g = -0.589$). Post-treatment recovery speed ($\mu_\kappa = -4.957$) implies a half-life of approximately 99 days. The interaction between treatment timing and recovery ($\beta_\kappa T = -0.569$) is negative, indicating that earlier treatment is associated with faster recovery. This is evidence that early funding accelerates

treatment timing, which generates both earlier access to care and faster improvement in health post-treatment.

Treatment lifts vary substantially across disease pathways (Online Appendix Table A6), ranging from 0.515 (Other Unclassified, Adult) to 2.352 (Organ Transplant, Adult). Transplant and HSCT segments consistently exhibit the largest treatment gains (lift above 1.6), reflecting the transformative nature of these interventions. Surgical segments show moderate lifts (1.1–1.5). Pre-treatment health is lowest for Organ Transplant Infants (2.205) and highest for Cardiac Surgery Adults (2.697), consistent with the clinical severity gradient from complex transplant cases to elective surgical procedures. Notably, the largest treatment lift accrues to Organ Transplant Adults (2.352), whose donor appeal is substantially lower than its pediatric counterparts in the same pathway, foreshadowing the misalignment that the counterfactual analysis quantifies.

Counterfactual Policy Simulations: Allocation based on different policies

With all three links of the M2H chain estimated, we can compute campaign-specific HROM and evaluate whether alternative allocation policies improve patient health. We evaluate the allocation policies via forward simulation of the full three-link system over the complete portfolio of 1,280 campaigns (of which 1,053 activate paid marketing within the observation window). The simulation uses parameters estimated on the 1,017-campaign estimation sample and preserves both the platform's realized daily budgets and the contemporaneous set of competing campaigns. The simulation procedure evolves recursively over calendar time: for each campaign, lagged donations, cumulative donations, and cumulative spend at the start of day t are those implied by the same policy through day $t - 1$. As a result, current allocations influence both contemporaneous donations and the state variables governing future outcomes. Donations are predicted using the

structural donation model (Equation 2), with the 2SRI control-function residual set to zero, thereby isolating exogenous variation in spending allocations.

The counterfactual environment holds fixed the observed aggregate daily budget and treats activation timing as exogenous. Prior to activation, campaigns receive zero paid spend and retain observed organic donations; on the activation date, spending is fixed at its observed value. Campaigns become eligible for reallocation beginning the day after activation. On each date with a positive budget, we solve the allocation problem directly over eligible campaigns' expenditures. Specifically, we compute (i) the donation-maximizing benchmark, (ii) the health-maximizing allocation, and (iii) for each retention target r , the health-maximizing allocation subject to achieving at least a fraction r of the donations obtained under the donation-maximizing benchmark. All allocations are characterized by their equalized marginal conditions rather than by imposing a parametric adjustment path.

Table 6 reports outcomes for the donation-maximizing benchmark (Don-Max), the donation-retention policies, and the health-maximizing allocation (Health-Max) benchmark, each evaluated holding the observed daily budget fixed. Figure 8 presents the implied donation–health frontier.

Table 6. Counterfactual Policy Outcomes (Full Portfolio; Donation-Maximizing Baseline)

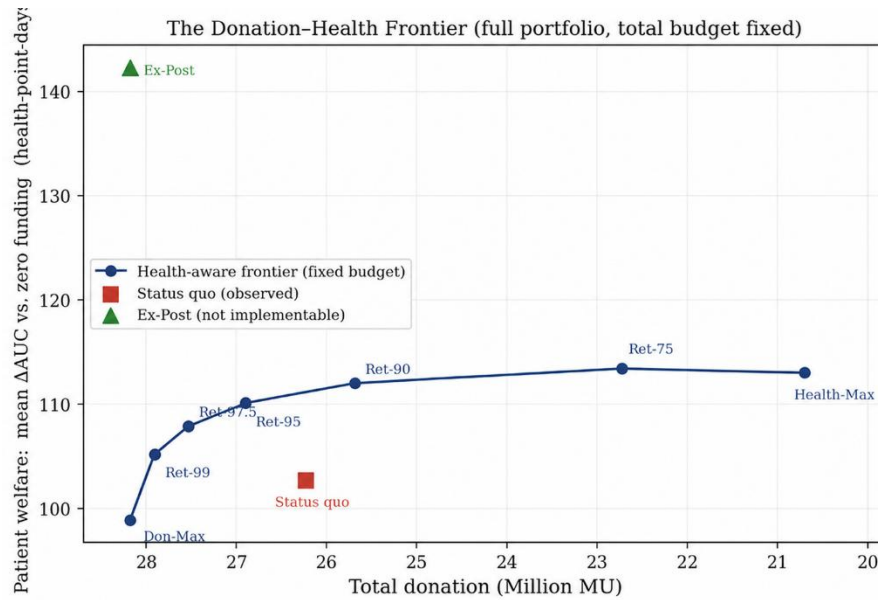
	Status Quo	Don- Max	Ret- 97.5	Ret- 95	Ret- 90	Ret- 75	Health- Max
<i>Level Metric</i>							
Total spend (million MU)	8.80	8.80	8.80	8.80	8.80	8.80	8.80
Total donations (million MU)	26.23	28.18	27.53	26.90	25.69	22.73	20.70
Cost per donation MU	0.335	0.312	0.319	0.327	0.342	0.387	0.425
Reaching 50% of goal (%)	61.7	58.4	63.3	63.9	64.5	64.7	64.1
Median fund. progress (%)	74.3	70.4	90.5	99.3	101.2	101.1	100.3
Δ AUC vs 0% funding	102.7	98.9	107.9	110.1	112.0	113.4	113.0
Treated ≤ 90 days (%)	32.0	29.6	34.5	34.8	35.5	35.9	35.9
Median treatment (days)	140.9	149.2	135.6	132.0	127.7	124.6	124.3

	Status Quo	Don- Max	Ret- 97.5	Ret- 95	Ret- 90	Ret- 75	Health- Max
<i>Comparison vs Donation-Maximizing Baseline</i>							
Donation change vs DM (%)	-6.9	--	-2.3	-4.5	-8.8	-19.3	-26.6
Health change vs DM (%)	+3.8	--	+9.1	+11.3	+13.2	+14.6	+14.2
Δ AUC vs DM (per campaign)	+3.7	--	+9.0	+11.2	+13.0	+14.4	+14.1
AUC gain / million MU of don. lost	2,452	--	17,597	11,218	6,702	3,390	2,403

Notes: All policies preserve the observed daily marketing budget across the full sample (N = 1,280 campaigns). Status Quo uses observed campaign-day marketing allocation and donations in MU. Don-Max solves the daily first-order conditions to maximize predicted donations and serves as the comparison baseline. Ret-X retains at least X% of Don-Max predicted donations on each calendar date while maximizing health gain by directly optimizing over all eligible campaign spends; Health-Max maximizes health with no donation floor. All policies set $\hat{v} = 0$ and use recursive state updates. Δ AUC is mean per-campaign health gain relative to zero funding (365-day horizon, joint AFT–OU model). Outcomes are reported for the full 1,280-campaign portfolio, of which 1,053 campaigns activate paid marketing within the modeled window.

The donation–health frontier is concave (Figure 8). The full extent of the misalignment is substantial: at the health-maximizing end of the frontier, patient health improves by up to 14.6% higher than under donation maximization (Ret-75), though at a considerable 19.3% donation cost. However, the concavity of the frontier implies that most of this health is recoverable at far more modest cost. At the conservative end, Ret-97.5 sacrifices only 2.3% of donation-maximizing fundraising while raising patient health by 9.1% (Table 6), with a cost per donation MU (0.319) still below the status quo (0.335). At Ret-95, health gains reach 11.3% at a 4.5% donation sacrifice, illustrating the diminishing marginal returns of further donation sacrifice visible in the frontier's flattening slope.

Figure 8: The Donation–Health Frontier

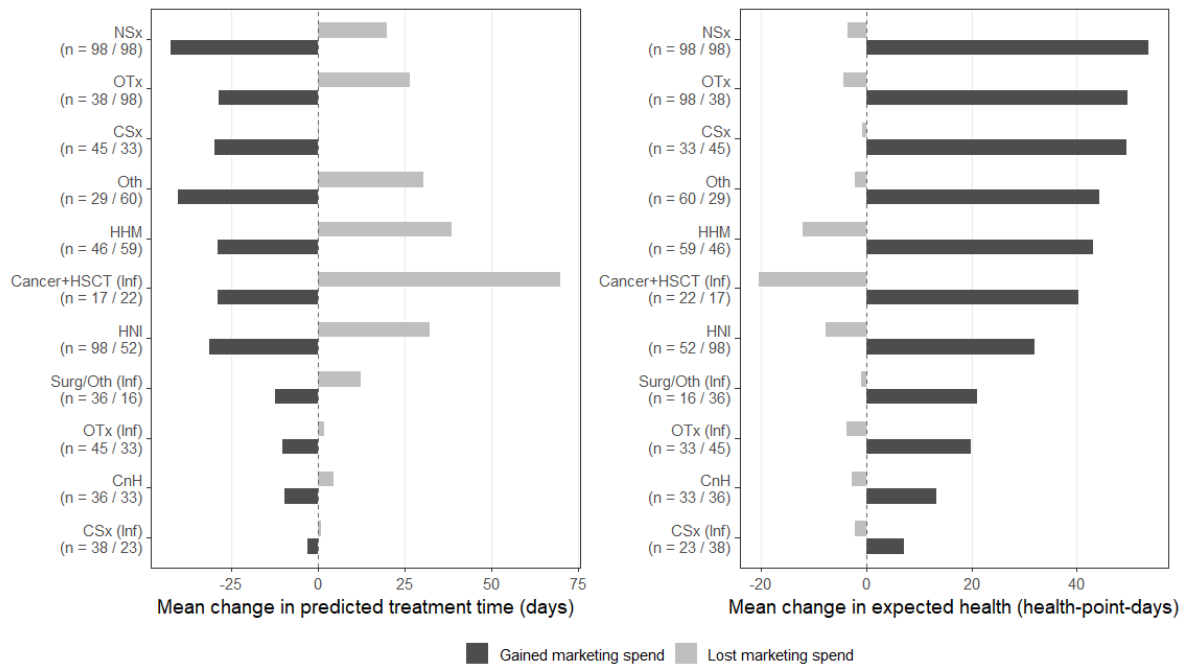


Notes: Each marker is an allocation policy from Table 6, holding total marketing spend fixed at its observed level. The horizontal axis is total donations (Million MU, decreasing rightward); the vertical axis is mean patient health (ΔAUC over a 365-day horizon relative to zero early funding). The blue curve is the health-aware frontier traced by the donation-retention level r , from donation maximization ($r = 1$) to the unconstrained health-maximizing allocation (Health-Max). The status quo lies inside the frontier.

The health gains operate through the funding–timing channel established in Links 2 and 3. Median funding progress rises from 70.4% under donation maximization to 90.5% under Ret-97.5 and 101.1% under Ret-75. The share of patients predicted to receive treatment within 90 days rises from 29.6% to 34.5% under Ret-97.5 and 35.9% under Ret-75, and median treatment time falls from 149.2 days to 135.6 under Ret-97.5 and 124.6 under Ret-75.

Figure 9 decomposes the reallocation into winners and losers by clinical pathway under Ret-75. The pattern is asymmetric in every segment: campaigns gaining marketing spend improve in health 3–10 times more than campaigns losing spend decline, averaging 9 to 40 health-point-days depending on the pathway, while the health loss averages just 0.2 to 15 health-point-days. Health-aware allocation therefore reallocates marketing spend from overfunded campaigns to underfunded campaigns with higher marginal health returns.

Figure 9: Segment-Level Health Gains and Losses (Ret-75 vs Donation-Maximizing)



Notes: Left panel: mean change in predicted treatment time (days). Right panel: mean change in expected health (health-point-days). Dark bars: campaigns that gained marketing spend under the health-aware policy. Light bars: campaigns that lost marketing spend. N labels show the count of campaigns gaining/losing per pathway.

7. DISCUSSION

The results reveal three key findings about the donation–health tradeoff in performance marketing for medical crowdfunding.

Performance marketing versus the health-maximizing policy: A donation-maximizing allocation rationally concentrates spend on high-appeal campaigns, where the intercept-driven donation baseline dominates the marginal return. However, the campaigns with the highest marginal donation returns are not the same as those with the highest marginal health return. The concave shape of the donation–health frontier (Figure 8) reflects the magnitude of misalignment between marketing effectiveness and health returns at the campaign level. This is consistent with evidence from lab and field studies that donor responsiveness to fundraising appeals is only weakly related to the underlying medical need that determines health returns to early liquidity (Small et

al. 2007; Sudhir et al. 2016; Cryder et al. 2017). This provides an opportunity for the platform to partially bridge the resulting gap in health by taking into account the downstream health impact in its marketing allocation. The first 5–10% of donations sacrificed relative to donation maximization captures the most exploitable wedge, campaigns whose health returns substantially exceed what their donation ranking would suggest. Beyond roughly 25% of donations sacrificed (Ret-75), the health frontier flattens, producing little incremental health benefit while substantially reducing total funds raised. A platform facing stronger misalignment would find the frontier steeper and the health gains from aggressive reallocation correspondingly larger.

Health asymmetry across pathways. Campaigns gaining marketing spend under the health-aware policy improve in health 3–10× more than campaigns losing spend decline, across every clinical pathway. This asymmetry arises because the reallocation moves spend from campaigns on the flat portion of the health curve to campaigns on the steep portion. This result speaks to a longstanding concern in the health economics and resource allocation literature about the tension between efficiency and distributional outcomes (Wagstaff 1991; Bertsimas et al. 2012). A central finding in that literature is that allocating health care resources by maximizing aggregate health outcomes tends to concentrate resources and benefits among a subset of recipients. In medical crowdfunding, where the status quo allocation is driven by donation performance, maximizing health subject to a donation-performance constraint actually reduces concentration in both funding progress and health outcomes relative to the donation-maximizing policy. Health-aware allocation therefore improves aggregate health without explicitly targeting measures of need or distributional outcomes. These gains arise endogenously from redirecting spend toward underfunded campaigns with higher marginal health returns.

8. CONCLUSION

This paper quantifies how platform marketing allocation decisions translate into patient health outcomes by structurally linking marketing spend, fundraising dynamics, treatment timing, and health trajectories. Using data from 1,280 medical crowdfunding campaigns, we estimate the Health Return on Marketing (HROM) and evaluate the health implications of performance-based allocation.

Two findings emerge. First, reallocating a fixed marketing budget toward campaigns with higher HROM yields substantial health gains. Marketing-attributable health increases by up to 14.6%, with a 9.1% gain achievable while forgoing only 2.3% of donation-maximizing fundraising. The health–fundraising frontier is concave: the first few percent of donations forgone capture most of the potential health gains, and consequential tradeoffs arise only at higher levels of health targeting. These gains arise because performance-based marketing allocation systematically prioritizes campaigns with high fundraising appeal rather than those where early funding most accelerates treatment and improves health. As a result, campaigns facing the largest financial constraints, exhibiting the lowest pre-treatment health, or delivering the greatest treatment-induced improvements are persistently underserved.

Second, the patient health benefits of reallocation are highly asymmetric. Campaigns that gain funding experience health improvements several times larger than the losses incurred by campaigns that lose funding, a pattern that holds across all clinical pathways. This asymmetry reflects the nonlinear funding-to-treatment relationship: early funding below the goal has a disproportionately large impact on treatment access, while additional funding beyond the goal yields diminishing health returns. Consequently, health-aware allocation redirects resources away from campaigns on flatter portions of the curve translating fundraising success to health

improvement and toward those for whom incremental marketing-induced fundraising progress yields greater incremental health value. A byproduct is that funding progress becomes more broadly distributed across campaigns, even though equity is not an explicit objective.

These results contribute to multiple streams of marketing research. First, to research on the impact of marketing on society, we move beyond marketing and its immediate consumer responses by tracing how platform marketing decisions impact donations, patient treatment timing and health outcomes, operationalized through the M2H chain. Second, the findings speak to the platform design literature, by showing that marketing allocation based on donation performance in medical crowdfunding generates systematic misalignment between platform decisions and beneficiary outcomes, highlighting the importance of accounting for outcomes on both sides of the market. Third, to marketing resource allocation, we develop HROM as a measure of patient health benefits and show that health-aware marketing allocation policies can recover most of the foregone health benefits, compared to the donation-maximizing policy, with minimal loss in fundraising performance.

For platform managers, the results suggest that health-improving allocation is operationally feasible. The key inputs required to approximate HROM, including clinical pathway, fundraising goal, and hospital identity are observable at campaign intake, and coarse segment-level estimates capture most of the available gains. Managers need not abandon fundraising performance as an objective. Instead, donation-retention constraints can be used to identify allocations that improve patient health while preserving nearly all of the fundraising generated under a donation-maximizing policy.

Several limitations point to directions for future research. First, the analysis holds donor response behavior fixed; if changes in allocation alter donor engagement or campaign selection behavior, realized outcomes may differ. Understanding these broader equilibrium responses is an important

area for field experimentation. A related question is whether advertising content that communicates downstream health impact can shift donor responsiveness toward high-return campaigns, bridging the misalignment between donation productivity and health returns within the existing earmarked structure. Critically, whether it can do so without reducing total donations, or whether informing donors about health impact dampens giving overall even as it redirects it, remains an open empirical question. Second, the results are most applicable to settings where out-of-pocket costs constrain treatment access; in environments with comprehensive insurance, the funding-to-treatment link would be weaker. Third, improving allocation decisions would benefit from richer clinical information at intake, enabling more granular estimation of campaign-level health returns. The empirical framework is modular and can incorporate such improvements without structural changes. Finally, the counterfactual policies optimize allocations one day at a time conditional on beginning-of-day states and then update those states recursively. A fully forward-looking policy could additionally internalize how current spending affects future fundraising states. The reported frontier may therefore conservatively characterize the health gains attainable under a broader class of intertemporal policies.

More broadly, in donation-based crowdfunding, donation-maximizing allocation can systematically direct resources towards popular campaigns at the cost of downstream beneficiary outcomes when donor responsiveness is misaligned with beneficiary need. Incorporating downstream health into allocation decisions can therefore transform marketing from a tool for maximizing engagement into a mechanism for improving real-world outcomes, often without materially compromising fundraising performance.

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